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CHOI et al.

(54) **DISPLAY DEVICE**

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(57) **ABSTRACT**

According to an embodiment, a display device may include a display panel including a display area and a non-display region disposed outside the display area, and including a flexible substrate including a protruded area extended from a portion of the non-display area, and an antenna circuit board connected to the protruded area. The non-display region includes a first non-display area positioned on a long side of the display panel, a second non-display area positioned on a short side of the display panel and a third non-display area positioned on a corner of the display panel, connecting the first non-display area with the second non-display area, and the display panel includes at least one transmitting antenna disposed at least partially in, or adjacent to, the first non-display area, adjacent first receiving antennas and a second receiving antenna disposed in the third non-display area.

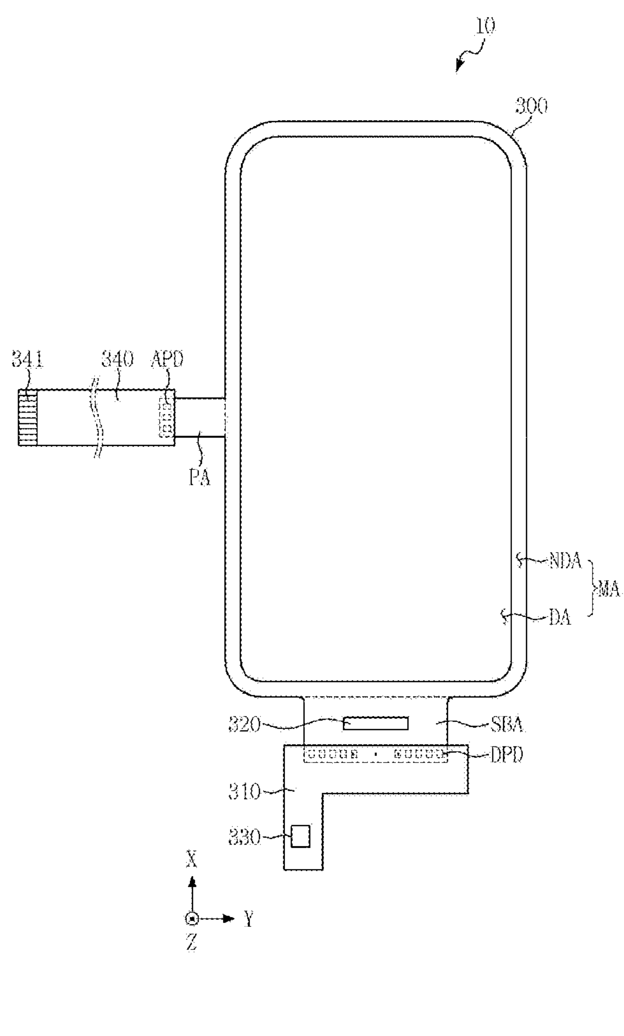


FIG. 1

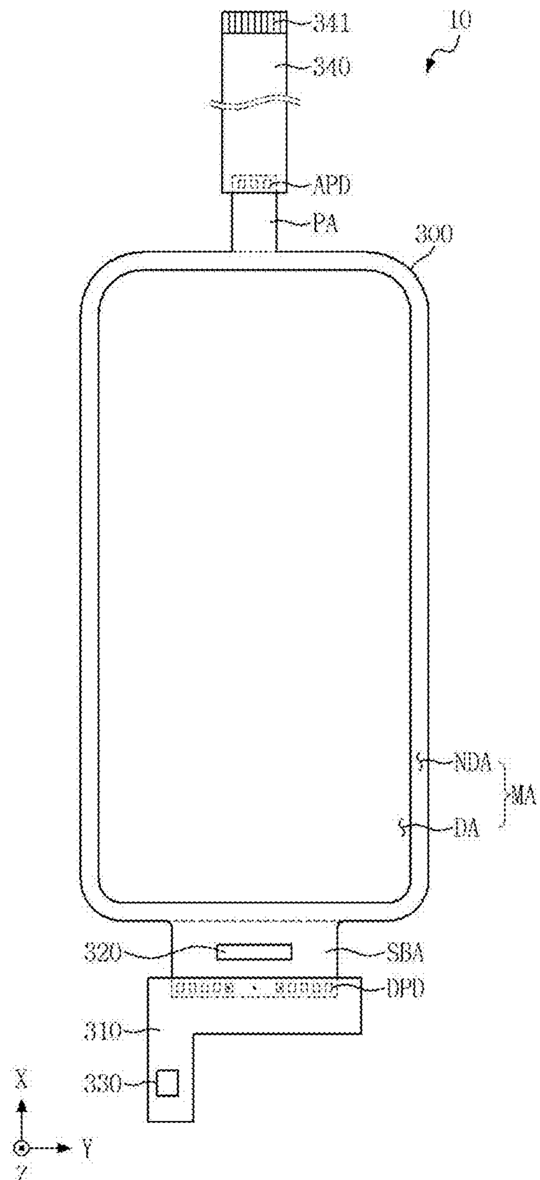


FIG. 2

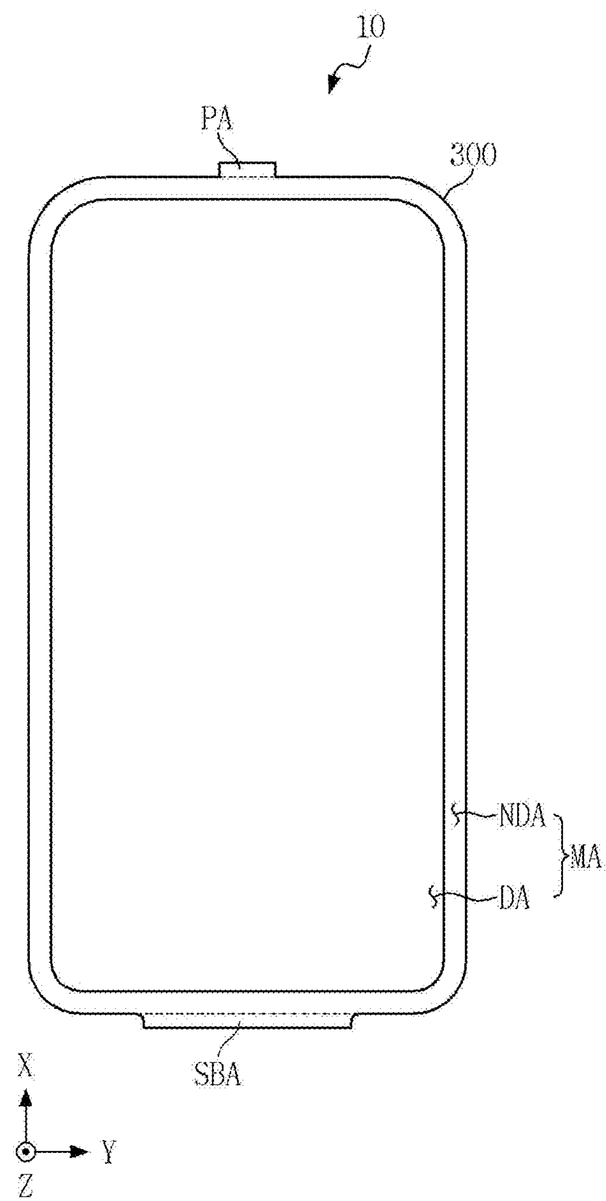
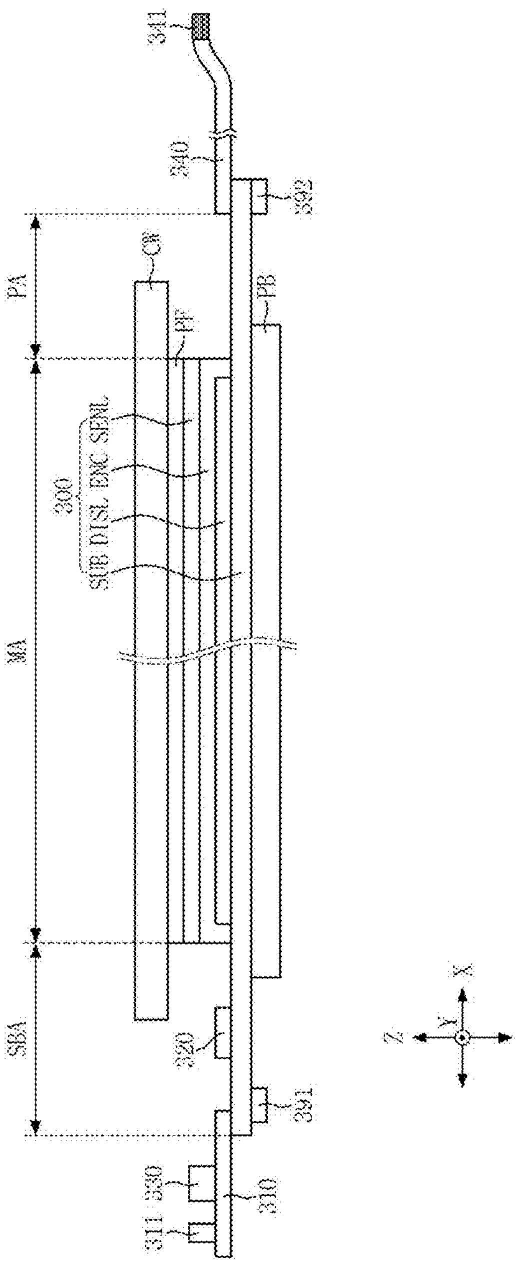


FIG. 3


341
340
392
PA
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SUB DISL ENC SENL
MA
SBA
320
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311 330
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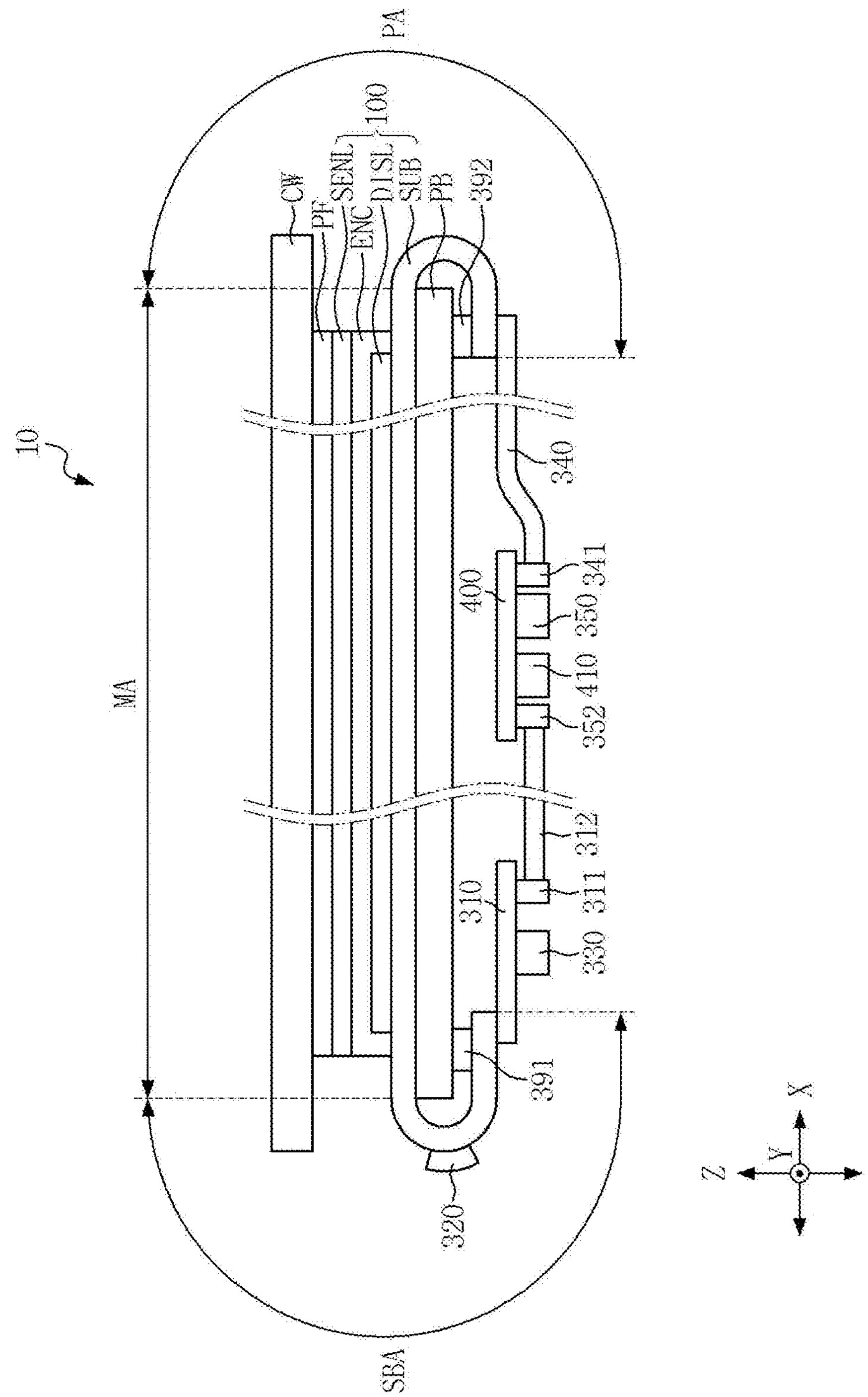
X
Y
Z

FIG. 4



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ENC
DISL
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PB
392
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MA
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SBA
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Y
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FIG. 5

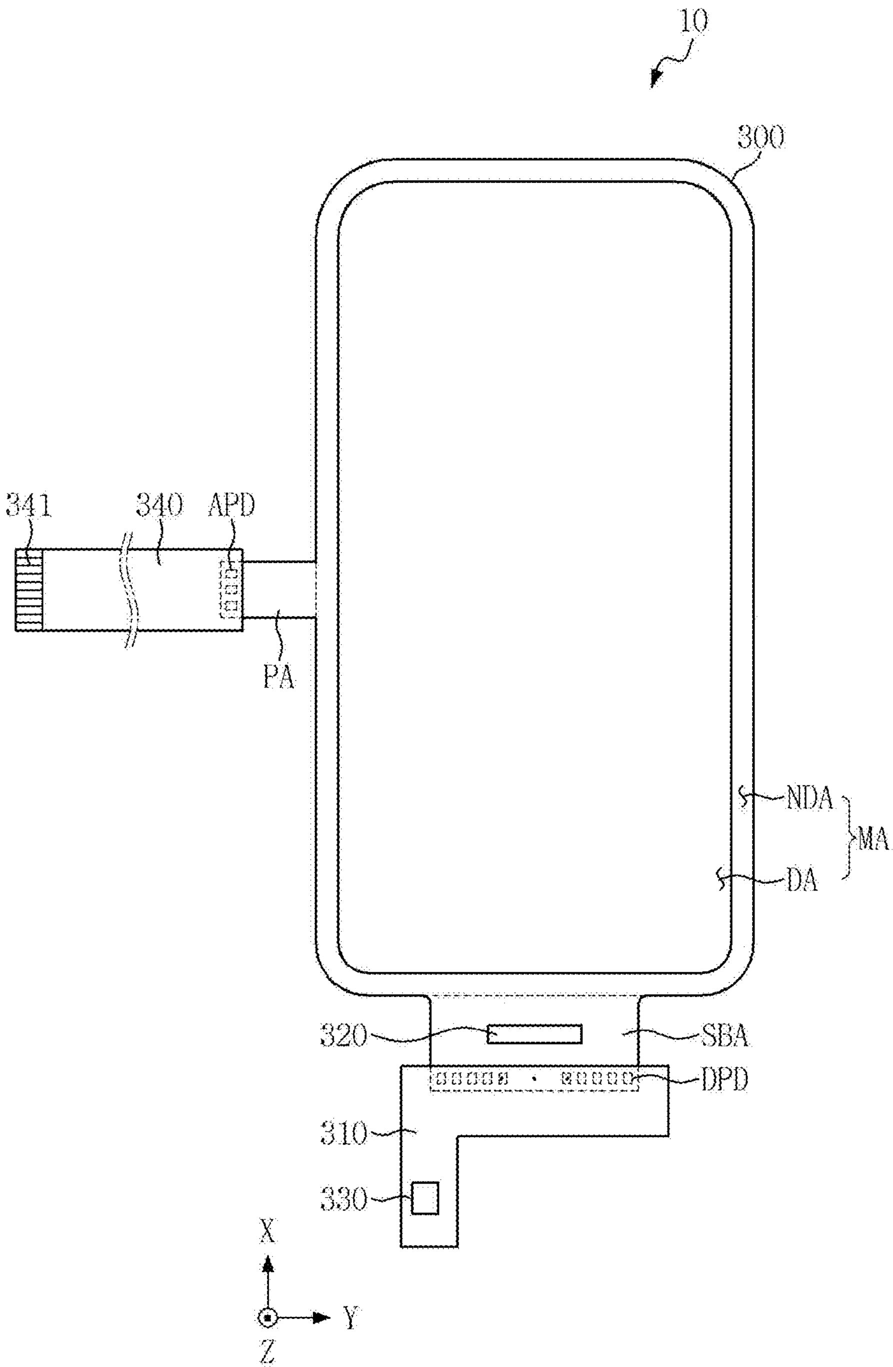
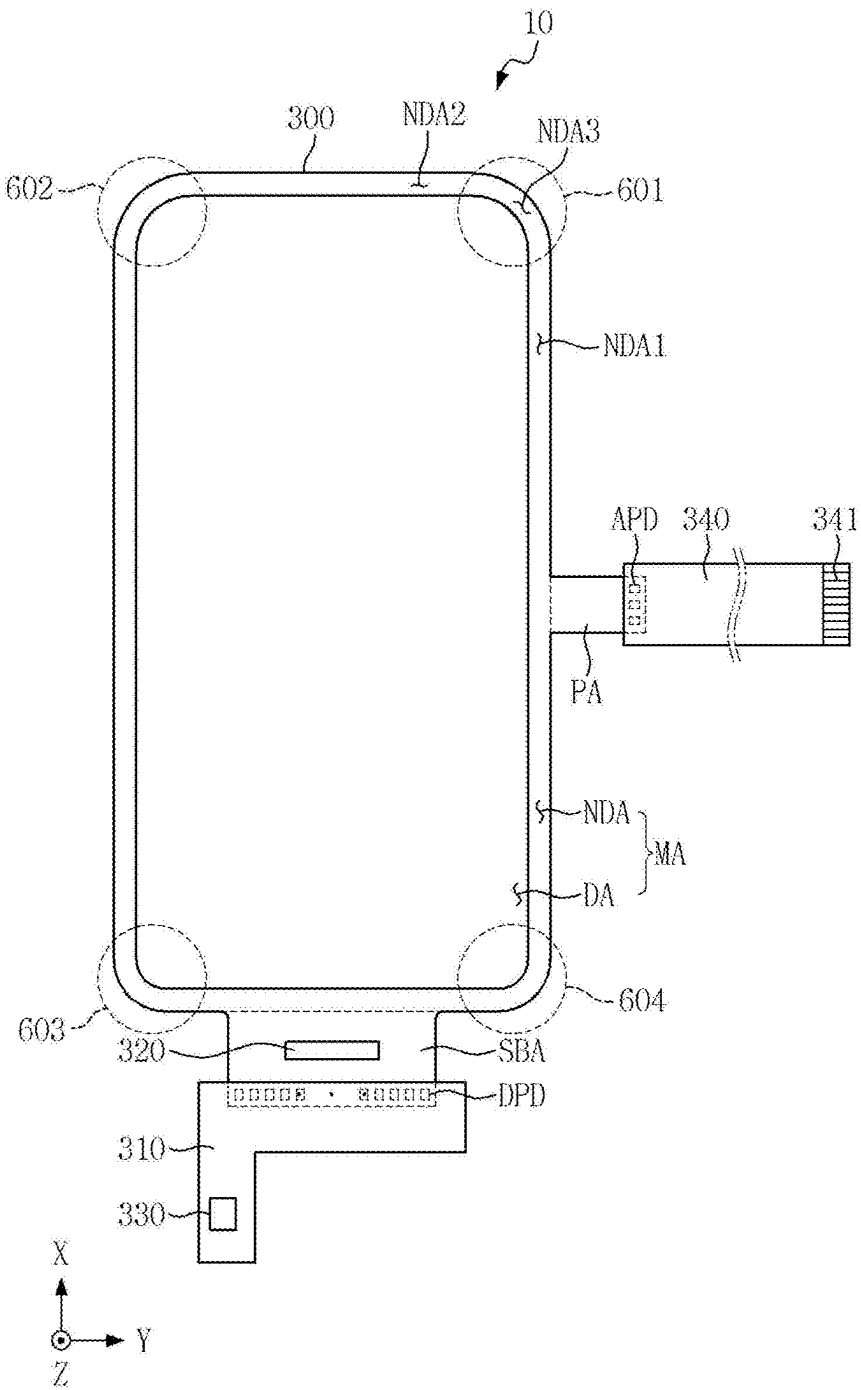


FIG. 6



10
300
NDA2
NDA3
602
601
NDA1
APD
340
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NDA
MA
DA
603
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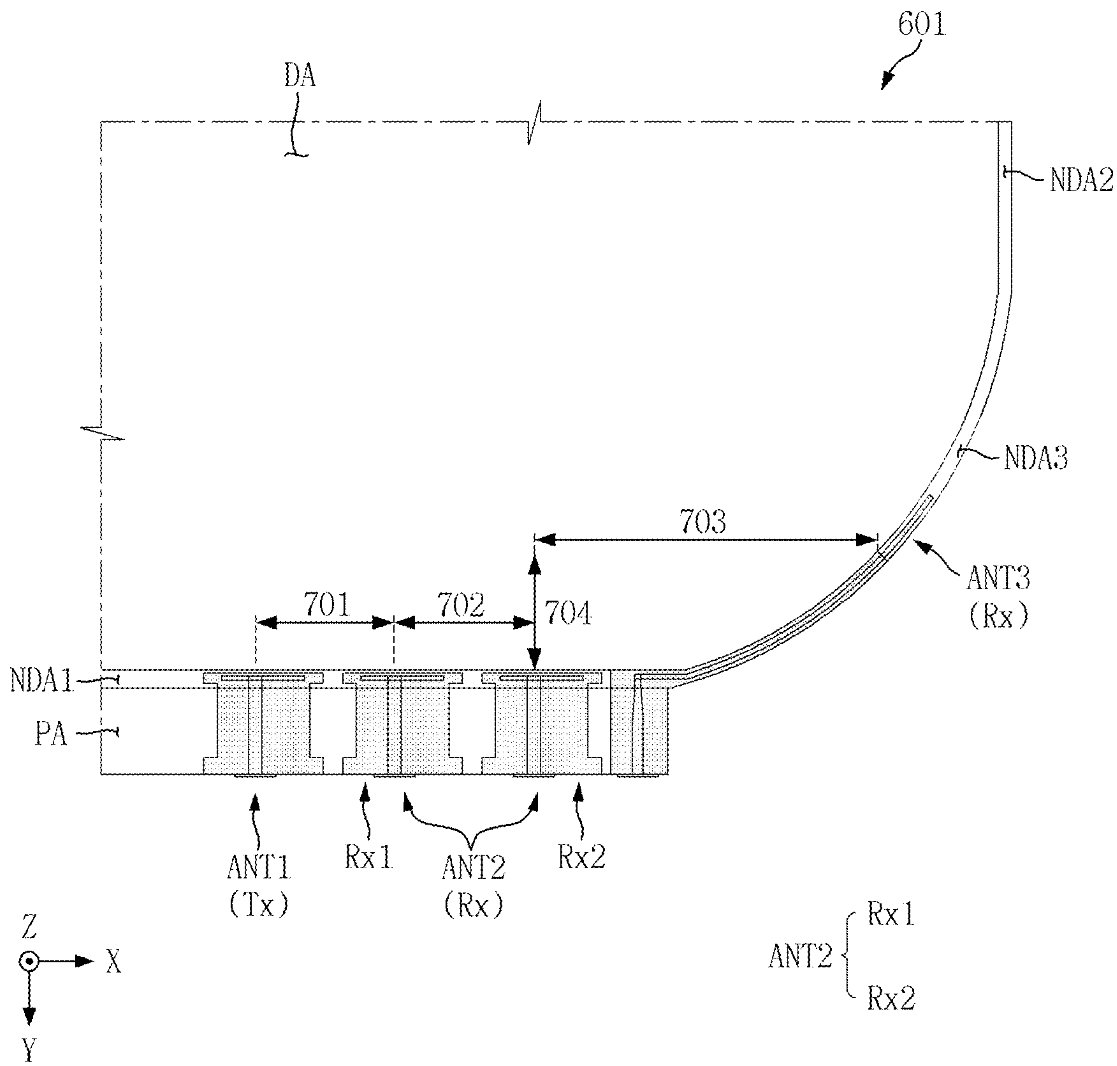
DPD
310
330
X
Y
Z

FIG. 7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ANT2
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Rx2
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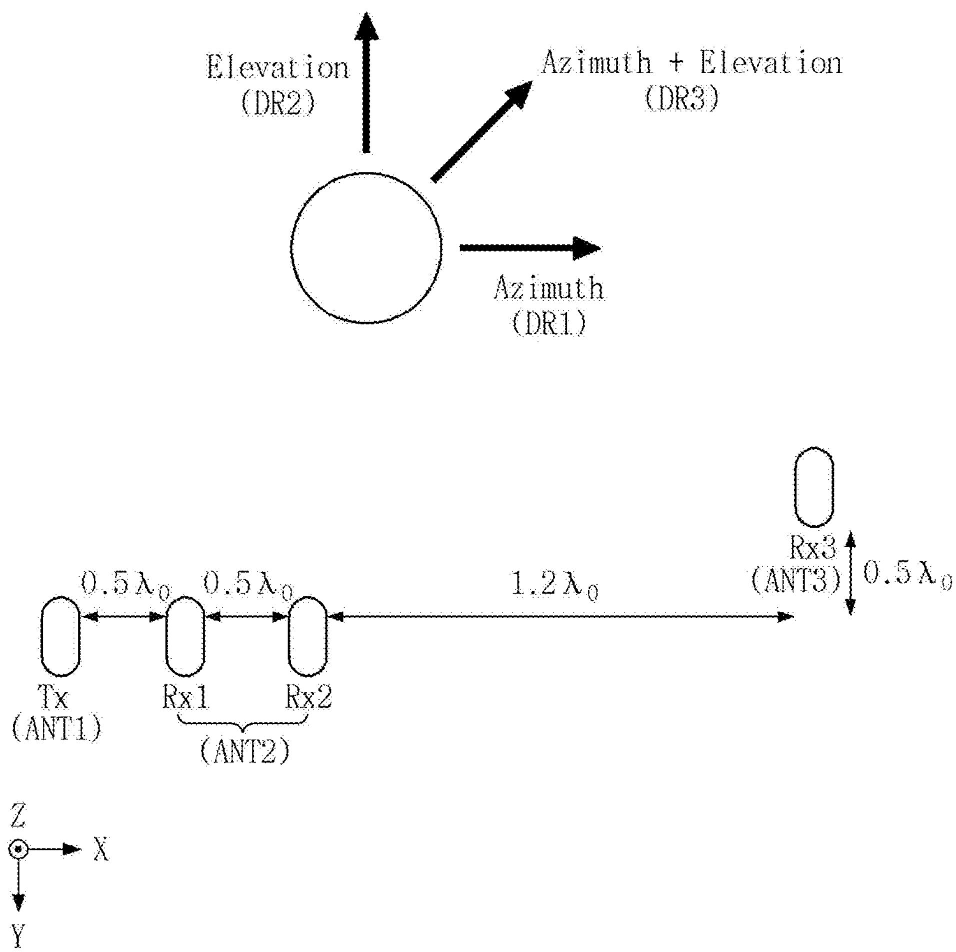
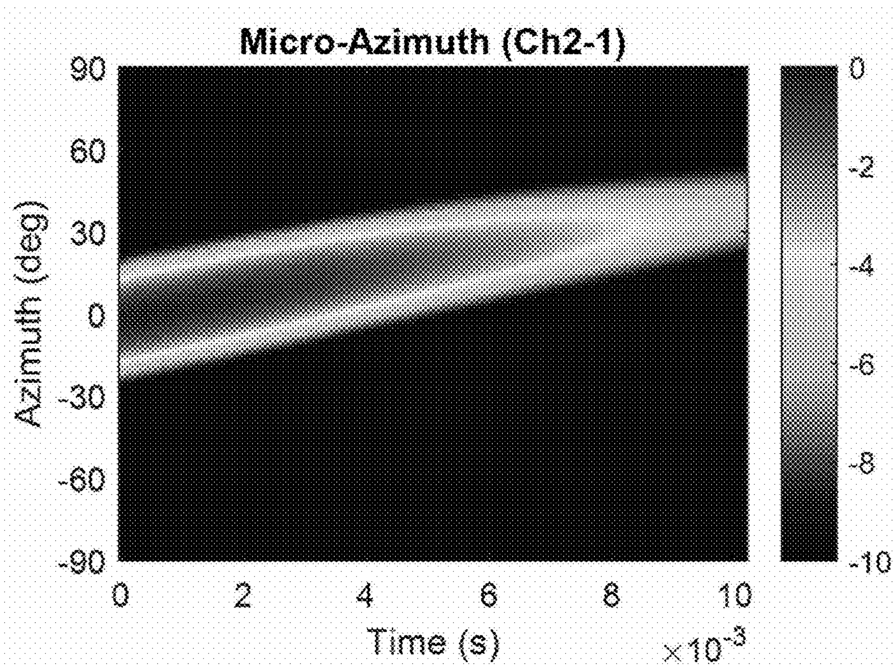
ANT2
Rx1
Rx2

FIG. 8



Elevation
(DR2)
Azimuth + Elevation
(DR3)
Azimuth
(DR1)
Rx3
(ANT3)
0.5λ0
0.5λ0
0.5λ0
1.2λ0
Tx
(ANT1)
Rx1
Rx2
(ANT2)
Z
X
Y

FIG. 9A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Azimuth (deg)
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-60
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10
Time (s)
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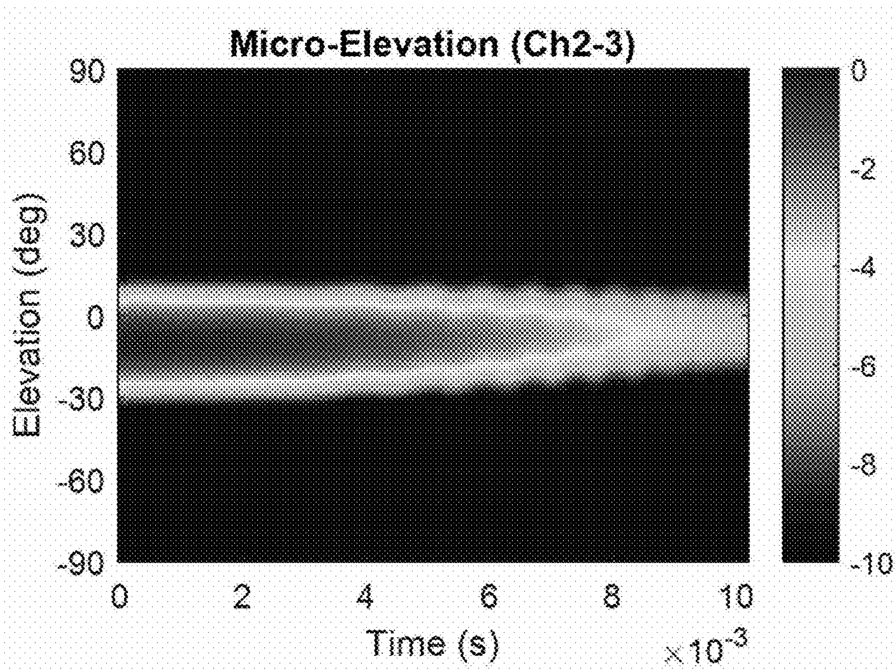
-4
-6
-8
-10

FIG. 9B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Time (s)
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-4
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-8
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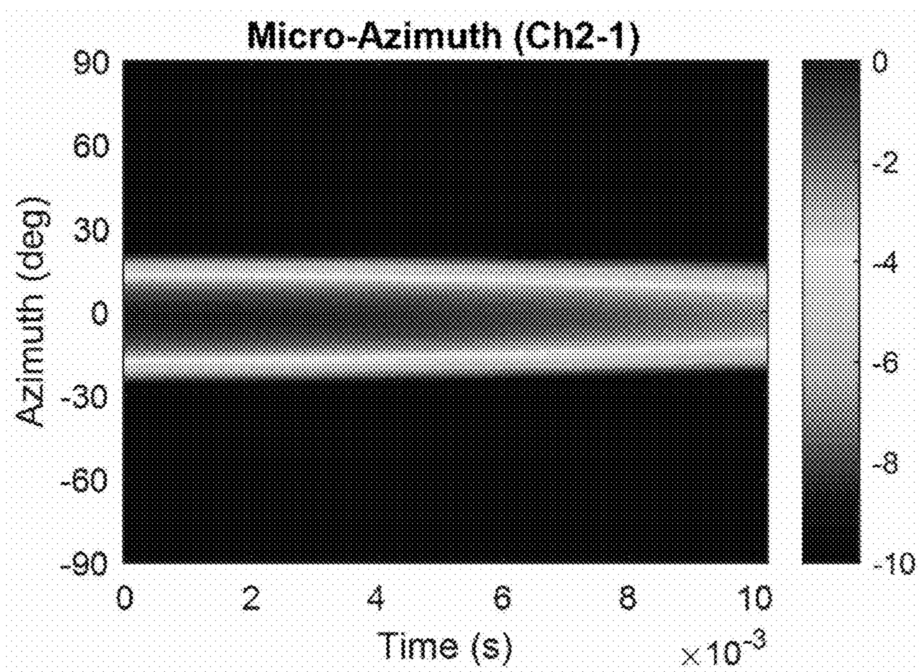
FIG. 10A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Azimuth (deg)
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Time (s)
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×10^-3
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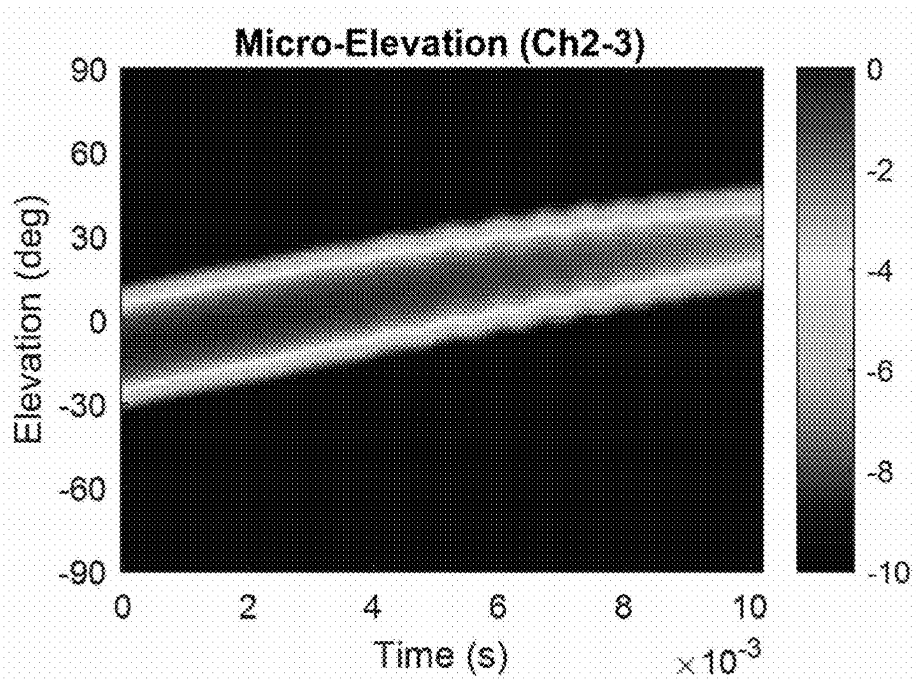
-4
-6
-8
-10

FIG. 10B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Time (s)
$\times 10^{-3}$
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-4
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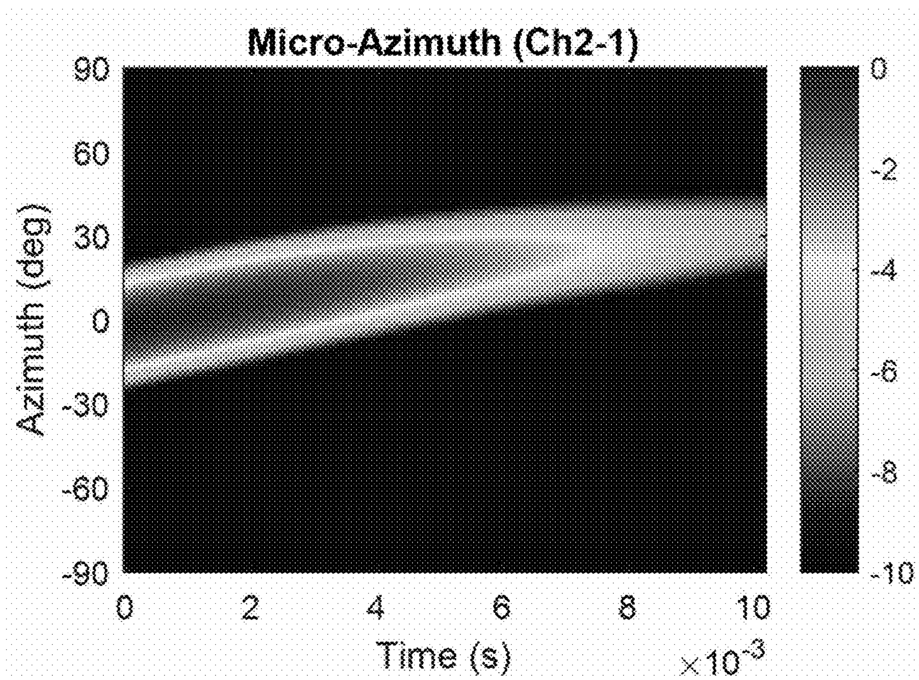
-8
-10

FIG. 11A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Azimuth (deg)
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-60
-90
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Time (s)
$\times 10^{-3}$
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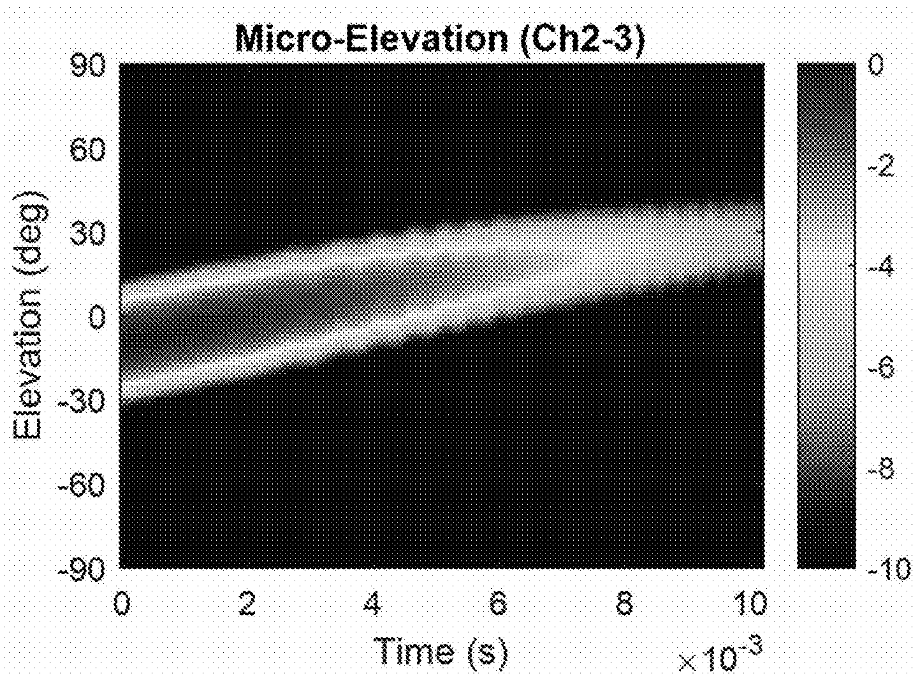
-4
-6
-8
-10

FIG. 11B


Micro-Elevation (Ch2-3)
Elevation (deg)
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-30
-60
-90
0
2
4
6
8
10
Time (s)
$\times 10^{-3}$
0
-2
-4
-6
-8
-10

FIG. 12

Name	X	Y
m1	104.6731	110.7714

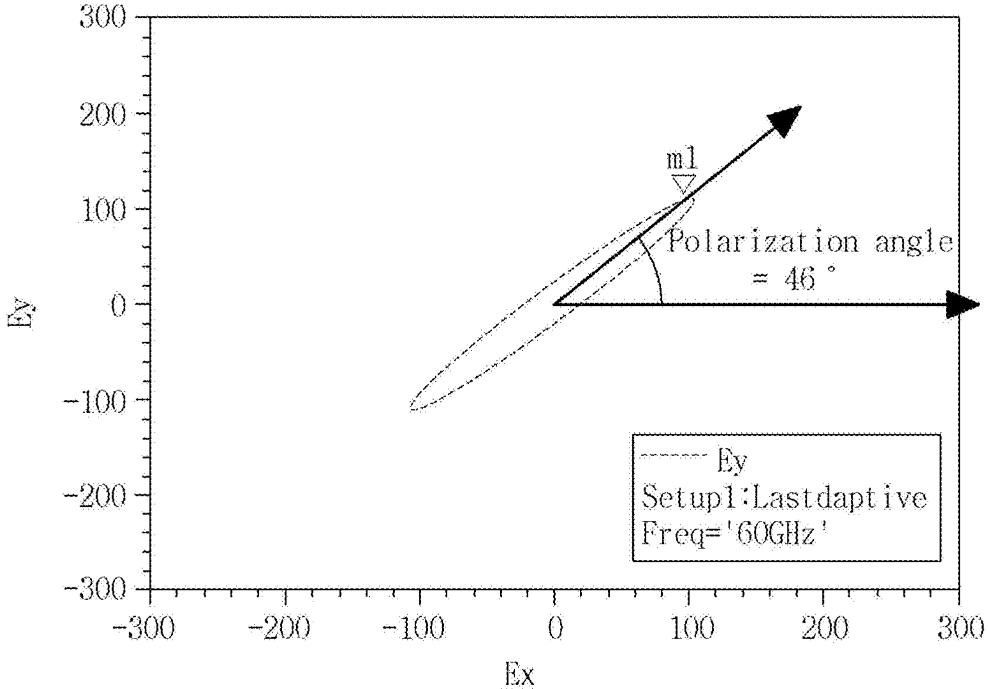
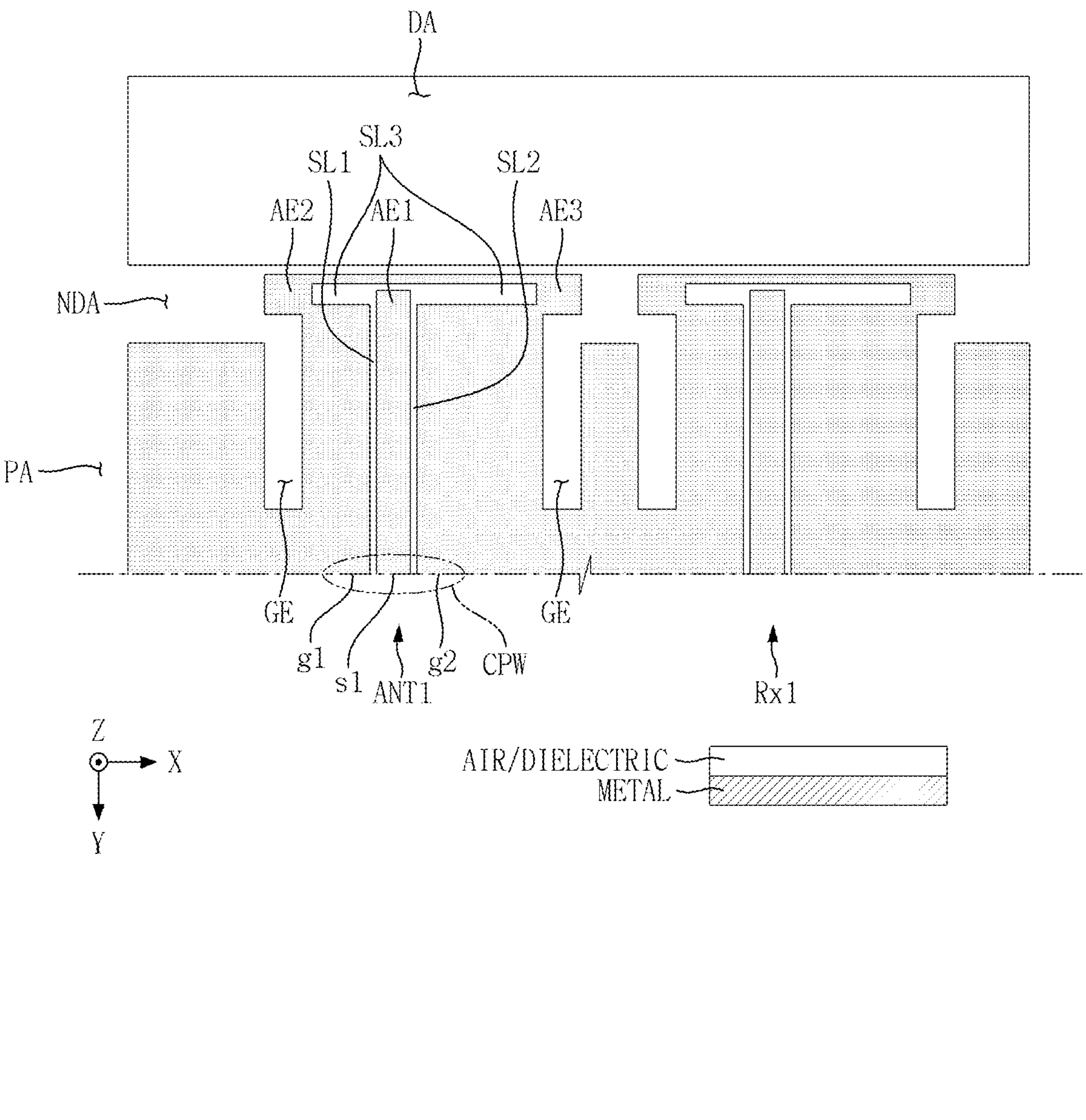


FIG. 13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SL1
SL2
AE2
AE1
AE3
NDA
PA
GE
GE
g1
s1
ANT1
g2
CPW
Rx1
Z
X
Y
AIR/DIELECTRIC
METAL

FIG. 14

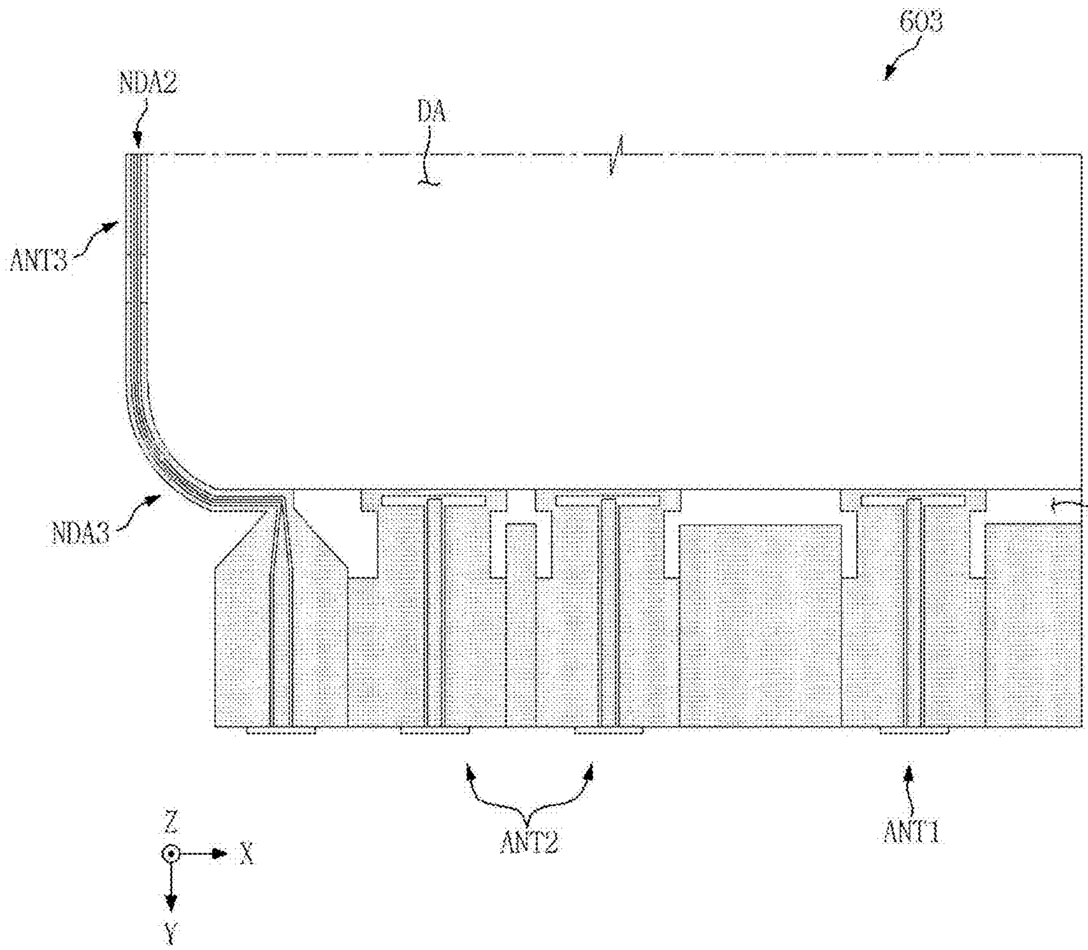
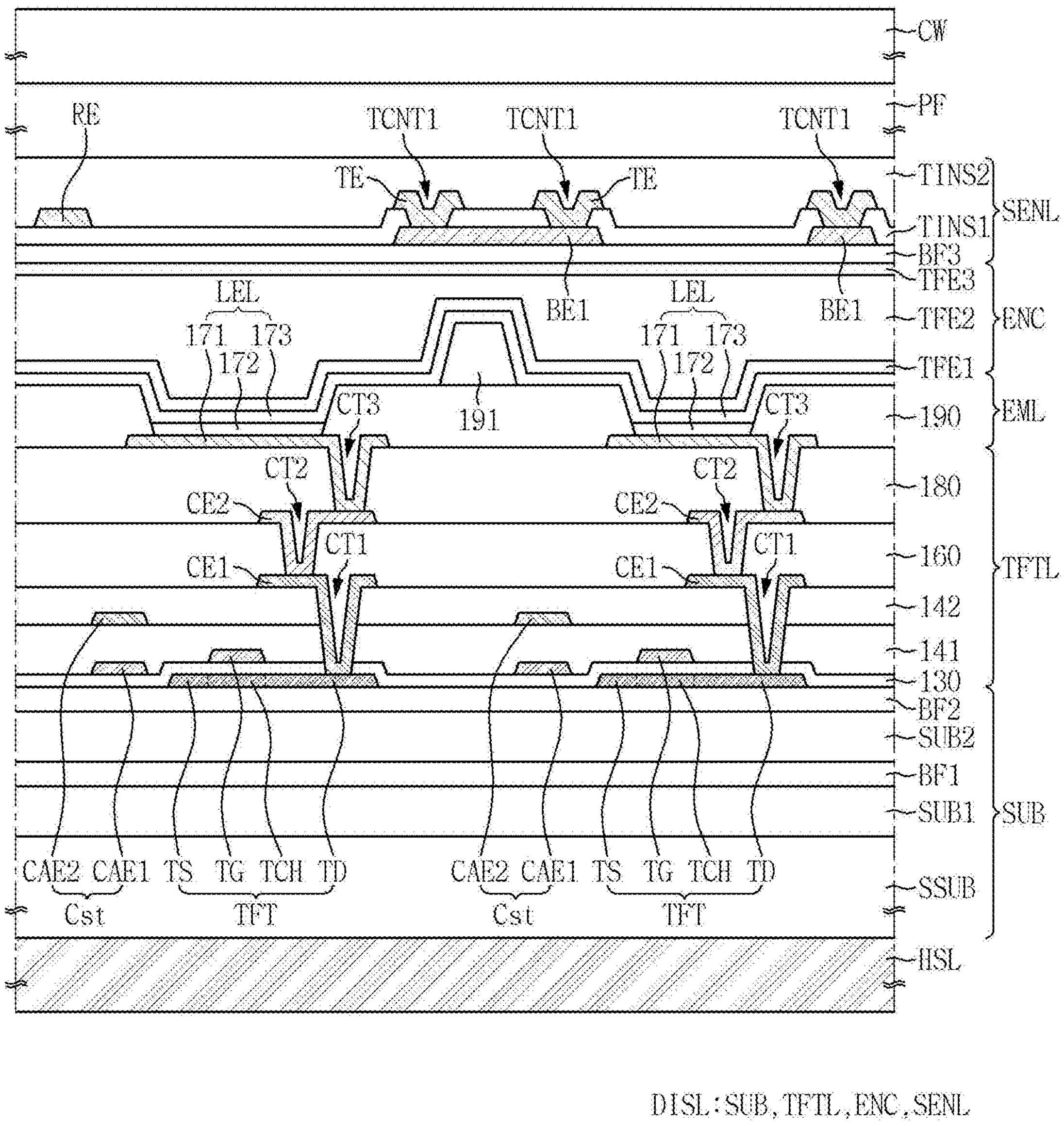


FIG. 15



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PF
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TCNT1
TCNT1
TE
TE
TINS2
TINS1
SENL
BF3
TFE3
BE1
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LEL
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172
171 173
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TFE2
ENC
TFE1
CT3
CT3
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EML
CT2
CT2
180
CE2
CE2
160
TFTL
CE1
CE1
CT1
CT1
142
141
130
BF2
SUB2
BF1
SUB1
SUB
CAE2 CAE1 TS TG TCH TD
CAE2 CAE1 TS TG TCH TD
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TFT
Cst
TFT
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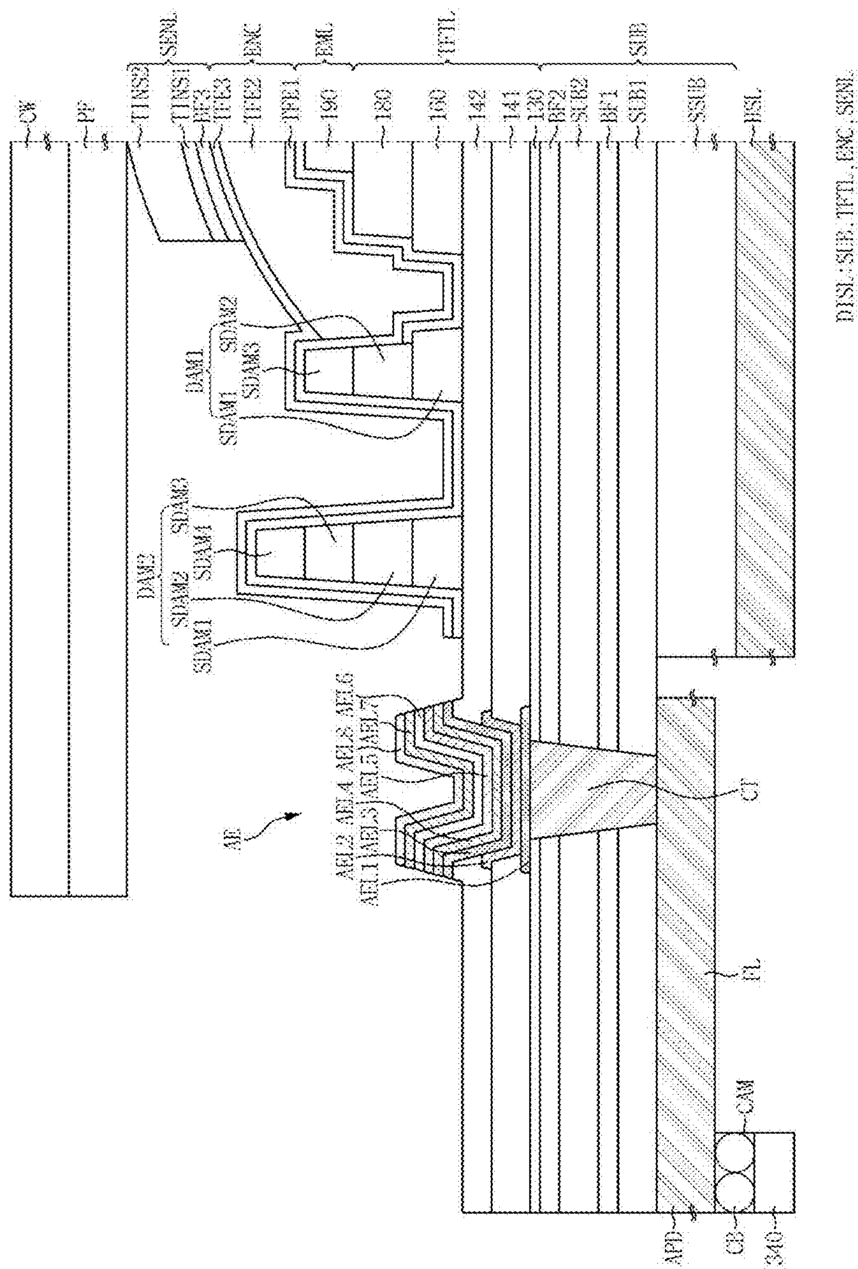
HSL
DISL:SUB,TFTL,ENC,SENL

FIG. 16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BF1
SUB1
SSUB
BSL
SENL
ENC
EML
TFTL
SUB
DAM1
SDAM1
SDAM2
SDAM3
DAM2
AE
AEL1
AEL2
AEL3
AEL4
AEL5
AEL6
AEL7
AEL8
CT
FL
CAM
APD
CB
340
DISL: SUB, TFTL, ENC, SENL

DISPLAY DEVICE

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application claims priority from Korean Patent Application No. 10-2024-0031157 filed on Mar. 5, 2024, in the Korean Intellectual Property Office and all the benefits accruing therefrom under 35 U.S.C. 119, the contents of which are incorporated by reference herein in its entirety.

TECHNICAL FIELD

[0002] The present disclosure relates generally to a display device and more particularly to a display device including embedded antennas.

DESCRIPTION OF THE RELATED ART

[0003] With recent advances in information technology, the demand for various types of display devices has expanded. Display devices are now prevalent in smart phones, tablets, laptops, digital cameras, vehicle navigators, and so forth.

[0004] A mobile electronic device may include an antenna for transmitting and receiving electromagnetic waves for wireless communication and/or various types of sensing. For example, the mobile electronic device may include a multi-input multi-output (MIMO) antenna for hand gesture recognition (HGR), which recognizes a user's hand gesture for receiving input commands. The MIMO antenna may include a plurality of transmitting antennas and a plurality of receiving antennas. The MIMO antennas may be disposed in a bezel of the mobile electronic device.

SUMMARY

[0005] Embodiments of the present disclosure relate to a mobile display device having a reduced bezel size, in which antennas serving as sensors are disposed in a non-display area rather than a bezel.

[0006] According to an embodiment of the disclosure, a display device may include a display panel including a display area and a non-display region disposed outside the display area, and including a flexible substrate including a protruded area extended from a portion of the non-display area, and an antenna circuit board connected to the protruded area. The non-display region includes a first non-display area along a long side of the display panel, a second non-display area along a short side of the display panel and a third non-display area positioned on a corner of the display panel, connecting the first non-display area with the second non-display area, and the display panel includes at least one transmitting antenna disposed at least partly in or adjacent to the first non-display area, a plurality of first receiving antennas adjacent to the at least one transmitting antenna and disposed at least partly in or adjacent to the first non-display area, and a second receiving antenna disposed in the third non-display area.

[0007] In various embodiments:

[0008] The at least one transmitting antenna includes a first antenna, and the plurality of first receiving antennas include a second antenna and a third antenna, which are disposed in parallel with each other.

[0009] The first to third antennas are the same type of antenna and have the same shape.

[0010] The shape of the first to third antennas is different from a shape of the second receiving antenna.

[0011] The first to third antennas are dipole antennas, and the second receiving antenna is a monopole antenna.

[0012] The first to third antennas are asymmetric dipoles having an asymmetric shape when viewed on a plane of the display panel.

[0013] Each of the first to third antennas includes a first antenna electrode extended from a feed line, a second antenna electrode extended in one side direction from the first antenna electrode, having a first length, and a third antenna electrode extended from the second antenna electrode in the other side direction opposite to the one side direction, having a second length.

[0014] A first ground pattern connected to the second antenna electrode is disposed in the one side direction from the feeding line, the first ground pattern including a first ground etched area disposed in parallel with the feeding line, and a second ground pattern connected to the third antenna electrode is disposed in the other side direction from the feeding line, the second ground pattern including a second ground etched area disposed in parallel with the feeding line.

[0015] According to an embodiment of the disclosure, a display device may include a display panel including a display area and a non-display region disposed outside the display area, and including a flexible substrate including a protruded area extended from a portion of the non-display area, and an antenna circuit board connected to the protruded area. The non-display region includes a first non-display area positioned on a short side of the display panel, a second non-display area positioned on a long side of the display panel and a third non-display area positioned on a corner of the display panel, connecting the first non-display area with the second non-display area, and the display panel includes at least one transmitting antenna disposed in the first non-display area, a plurality of first receiving antennas adjacent to the at least one transmitting antenna in the first non-display area, and a second receiving antenna disposed in the third non-display area.

[0016] According to an embodiment of the disclosure, a display device may include a display panel including a display area and a non-display region disposed outside the display area, and including a flexible substrate including a protruded area extended from a portion of the non-display area, and an antenna circuit board connected to the protruded area. The non-display region includes a first non-display area positioned on a long side of the display panel, a second non-display area positioned on a short side of the display panel and a third non-display area positioned on a corner of the display panel, connecting the first non-display area with the second non-display area, and the display panel includes at least one transmitting antenna disposed in the first non-display area, a plurality of first receiving antennas adjacent to the at least one transmitting antenna in the first non-display area, and a second receiving antenna disposed in the second non-display area.

[0017] A polarization direction of each of the first to third antennas is the same as a polarization direction of the second receiving antenna.

[0018] According to an embodiment of the disclosure, a display device may include a display panel including a display area and a non-display region disposed outside the display area, the display panel including a flexible substrate including a protruded area extended from a portion of the

non-display region, the display panel having a first side and a second side orthogonal to the first side; and an antenna circuit board connected to the protruded area. The non-display region includes a first non-display area along the first side of the display panel, a second non-display area along the second side of the display panel and a third non-display area on a corner of the display panel, connecting the first non-display area with the second non-display area, and the display panel includes at least one transmitting antenna disposed at least partly in or proximate to the first non-display area, a plurality of first receiving antennas disposed adjacent to the at least one transmitting antenna and at least partly in or proximate to the first non-display area, and a second receiving antenna disposed in the third non-display area.

[0019] In a display device according to the embodiments, antennas serving as sensors may be disposed in a non-display area, so that a bezel size of a mobile electronic device (within which the antennas may be otherwise disposed) may be reduced.

[0020] The effects according to the embodiments of the present disclosure are not limited to those mentioned above.

BRIEF DESCRIPTION OF THE DRAWINGS

[0021] The above and other aspects and features of the present disclosure will become more apparent by describing in detail exemplary embodiments thereof with reference to the attached drawings, in which:

[0022] FIGS. **1** and **2** are plan views illustrating a display device according to an embodiment;

[0023] FIG. **3** a cross-sectional view or side view depicting an example structure of a portion of the display device of FIGS. **1** and **2** with a protruded area PA and a sub-area SBA in unfolded states;

[0024] FIG. **4** is a cross-sectional view or side view depicting an example structure of the display device of FIGS. **1** and **2** with the protruded area PA and the sub-area SBA in folded states;

[0025] FIGS. **5** and **6** are plan views illustrating a display device according to another embodiment;

[0026] FIG. **7** is a plan view illustrating any one corner of a display panel according to an embodiment;

[0027] FIG. **8** is a conceptual view illustrating an operation of antennas according to an embodiment;

[0028] FIGS. **9**A and **9**B are radar images obtained by sensing an object, which is moving in a first direction, by antennas according to an embodiment;

[0029] FIGS. **10**A and **10**B are radar images obtained by sensing an object, which is moving in a second direction, by antennas according to an embodiment;

[0030] FIGS. **11**A and **11**B are radar images obtained by sensing an object, which is moving in a third direction, by antennas according to an embodiment;

[0031] FIG. **12** is a result of an experiment for a polarization direction of antennas according to an embodiment;

[0032] FIG. **13** is a plan view illustrating a transmitting antenna and a first receiving antenna according to an embodiment;

[0033] FIG. **14** is a plan view illustrating any one corner of a display panel according to one embodiment;

[0034] FIG. **15** is a cross-sectional view taken along a portion of a display area of a display panel according to an embodiment; and

[0035] FIG. **16** is a cross-sectional view taken along a boundary between an antenna area of a display panel and a non-display area adjacent thereto according to an embodiment.

DETAILED DESCRIPTION OF THE DISCLOSURE

[0036] Embodiments of the present inventive concept will now be described more fully hereinafter with reference to the accompanying drawings. The inventive concept may, however, be embodied in different forms and should not be construed as limited to the embodiments set forth herein. Rather, these embodiments are provided so that this disclosure will be thorough and complete, and will fully convey the scope of the claimed subject matter to those skilled in the art.

[0037] It will also be understood that when a layer is referred to as being "on" another layer or substrate, it can be directly on the other layer or substrate, or intervening layers may also be present. The same reference numbers indicate the same components throughout the specification.

[0038] It will be understood that, although the terms "first," "second," etc. may be used herein to describe various elements, these elements should not be limited by these terms. These terms are only used to distinguish one element from another element. For instance, a first element discussed below could be termed a second element without departing from the teachings of the present inventive concept. Similarly, the second element could also be termed the first element.

[0039] Features of each of various embodiments of the present disclosure may be partially or entirely combined with each other and may technically variously interwork with each other, and respective embodiments may be implemented independently of each other or may be implemented together in association with each other.

[0040] Hereinafter, specific embodiments will be described with reference to the accompanying drawings.

[0041] FIGS. **1** and **2** are plan views illustrating a display device according to an embodiment.

[0042] Referring to FIGS. **1** and **2**, a display device **10** according to an embodiment may be applied to a mobile electronic device such as a mobile phone, a smart phone, a tablet personal computer (PC), a mobile communication terminal, an electronic diary, an electronic book, a portable multimedia player (PMP), a navigator and an ultra mobile PC (UMPC). In other examples, the display device **10** may be applied to a television, a laptop computer, a monitor, a signboard or a display unit of Internet of things (IoT), or a wearable device such as a smart watch, a watch phone, an eyeglasses-type display and a head mounted display (HMD). In still other examples, the display device may be applied to a center information display (CID) disposed in a dashboard, a center fascia or a gauge of a vehicle, a room mirror display, a side mirror of a vehicle, or a display disposed on a rear surface of a front seat as an entertainment for a rear seat of a vehicle.

[0043] In the example, a display panel **300** of the display device **10** may have a plate-like geometry with an oblong profile in the XY plane (the plane parallel to the display surface) as illustrated. The oblong profile may be substantially rectangular with rounded corners, such that the display device **10** has a pair of parallel long sides which are orthogonal to a pair of parallel short sides. In describing some embodiments of the present disclosure, a first direction (X-axis direction, hereafter, "X direction") is a long side direction of the display device **10** and may be considered a vertical direction of the display device **10** when the display device is held vertically to present a "portrait" view. A second direction (Y-axis direction, hereafter, "Y direction") is a short side direction of the display device **10** and may be considered a horizontal direction of the display device **10** when the display device **10** is held to present a portrait view. A third direction (Z-axis direction, hereafter, "Z direction") may be a thickness direction of the display device **10**. A corner where a long side in the X direction) and a short side in the Y direction meet (are connected) may be rounded to have a predetermined curvature, such that the connected long and short sides are orthogonal.

[0044] The display device **10** may include the display panel **300**, a display circuit board **310**, a display driving circuit **320**, a touch driving circuit **330** and an antenna circuit board **340**. A connector **341** may be formed on one side of the antenna circuit board **340**.

[0045] The display panel **300** may be a light emitting display panel including a light emitting element. In other examples, the display panel **300** may be an organic light emitting display (OLED) panel using an organic light emitting diode including an organic light emitting layer; a micro light emitting diode display panel using a micro LED; a quantum dot light emitting display panel using a quantum dot light emitting diode including a quantum dot light emitting layer; or an inorganic light emitting display panel using an inorganic light emitting element including an inorganic semiconductor.

[0046] The display panel **300** may be a flexible display panel that is flexible and thus may be easily bent, folded or rolled. For example, the display panel **300** may be a foldable display panel that may be folded and unfolded; a curved display panel having a curved display surface; a bendable display panel in which an area other than a display surface is bent; a rollable display panel that may be rolled or unrolled; or a stretchable display panel that may be elongated.

[0047] The display panel **300** may include a main area MA, a sub-area SBA protruding from one side of the main area MA and a protruded area PA that protrudes from an opposite side of the main area MA.

[0048] The main area MA may include a display area DA for displaying an image and a non-display area (equivalently, "non-display region") NDA which is a peripheral area of the main area MA. The display area DA may occupy most of the main area MA. The display area DA may be disposed at the center of the main area MA. The non-display area NDA may be a bezel that surrounds the display area DA. The non-display area NDA may be defined as an edge area of the display panel **300**. The non-display area NDA may be referred to as "a dead space area".

[0049] The sub-area SBA may protrude from one side (e.g., the lower side in FIG. **1**) of the main area MA in the X. As shown in FIG. **1**, a length of the sub-area SBA in the X direction may be shorter than that of the main area MA in the X direction, and the length of the sub-area SBA in the Y direction may be shorter than that of the main area MA in the Y direction, but embodiments of the present disclosure are not limited thereto.

[0050] Referring to FIG. **2**, the sub-area SBA may be bent, and at least a portion of the bent sub-area SBA may be disposed below the display panel **300**. In this case, at least a portion of the sub-area SBA may overlap the main area MA of the display panel **300** in the Z direction.

[0051] Display pads DPD may be disposed at one edge of the sub-area SBA. One edge of the sub-area SBA may be a lower edge of the sub-area SBA. The display circuit board **310** may be attached to the display pads DPD of the sub-area SBA. The display circuit board **310** may be attached to the display pads DPD of the sub-area SBA by using a conductive adhesive member such as an anisotropic conductive film and an anisotropic conductive paste. The display circuit board **310** may be a flexible printed circuit board (FPCB) that may be bent, a rigid printed circuit board (PCB) that is rigid and thus is not easily bent, or a composite printed circuit board that includes both a rigid printed circuit board and a flexible printed circuit board.

[0052] The display driving circuit **320** may be disposed on the sub-area SBA of the display panel **300**. The display driving circuit **320** may receive control signals and power voltages, and may generate and output signals and voltages for driving the display panel **300**. The display driving circuit **320** may be formed of an integrated circuit (IC).

[0053] The touch driving circuit **330** may be disposed on (e.g., attached to) the display circuit board **310**. The touch driving circuit **330** may be formed of an integrated circuit.

[0054] The touch driving circuit **330** may be electrically connected to sensor electrodes of the sensor electrode layer of the display panel **300** through the display circuit board **310**. The touch driving circuit **330** may output a touch driving signal to each of the sensor electrodes and sense a voltage change according to mutual capacitance of the sensor electrodes.

[0055] The sensor electrode layer of the display panel **300** may sense a "proximity touch" and/or a "contact touch". Herein, "contact touch" means that an object, such as a person's finger or pen, comes into direct contact with a cover window disposed on the sensor electrode layer. Herein, "proximity touch" means that an object, such as a person's finger or pen, is positioned over the cover widow to be slightly spaced apart from the cover window, like hovering.

[0056] Display pixels of the display panel **300** and a power supply unit for supplying driving voltages for driving the display driving circuit **320** may be additionally disposed on the display circuit board **310**. Alternatively, the power supply unit may be integrated with the display driving circuit **320**, and in this case, the display driving circuit **320** and the power supply unit may be formed as one integrated circuit.

[0057] The protruded area PA may be an area that includes at least one element of a feeding line, a ground line or an antenna electrode of an antenna module for wireless communication. The protruded area PA may protrude from the opposite side of the main area MA in the X direction. For example, the opposite side of the main area MA may be an upper side of the main area MA. As shown in FIG. **1**, a length of the protruded area PA in the X direction may be shorter than that of the main area MA in the X direction, and a length of the protruded area PA in the Y direction may be shorter than that of the main area MA in the Y direction, but the embodiment of the present disclosure is not limited thereto.

[0058] As shown in FIG. **2**, at least a portion of the protruded area PA may be bent, and at least a portion of the bent protruded area PA may be disposed below the display panel **300**. In this case, at least a portion of the protruded area PA may overlap the main area MA of the display panel **300** in the Z direction.

[0059] Antenna pads APD may be disposed at one edge of the protruded area PA. The antenna circuit board **340** may be attached onto the antenna pads APD of the protruded area PA. The antenna circuit board **340** may be attached onto the antenna pads APD of the protruded area PA by using a conductive adhesive member such as an anisotropic conductive film and an anisotropic conductive adhesive. One side of the antenna circuit board **340** may include a connector **341** connected to a main circuit board **400** on which an antenna driving circuit (**350** of FIG. **4**) is packaged. The antenna circuit board **340** may be a flexible printed circuit board (FPCB).

[0060] FIG. **3** is a cross-sectional view or side view (depending on the opaqueness of the non-display area NDA) depicting an example structure of a portion of the display device **10** with the protruded area PA and the sub-area SBA in unfolded states. FIG. **4** is a cross-sectional view of aside view depicting an example structure of the display device **10** with the protruded area PA and the sub-area SBA in folded states. FIG. **4** shows additional example components of the display device **10** as compared to FIG. **4**.

[0061] Referring to FIGS. **3** and **4**, the display device **10** may include a display panel **300**, a polarizing film PF, a cover window CW, and a panel lower cover PB. The display panel **300** may include a substrate SUB, a display layer DISL, an encapsulation layer ENC and a sensor electrode layer SENL.

[0062] The substrate SUB may be made of an insulating material such as a polymer resin. The substrate SUB may be a flexible substrate capable of being subjected to bending, folding, rolling or the like.

[0063] In the main area MA, the display layer DISL may be disposed on the substrate SUB. The display layer DISL may be a layer that includes light emission areas to display an image. The display layer DISL may include a thin film transistor layer on which thin film transistors are formed, and a light emitting element layer in which light emitting elements for emitting light are disposed in the light emission areas.

[0064] Scan lines, data lines, power lines, etc. for driving the light emitting elements of the light emission area may be disposed in the display area DA of the display layer DISL. A scan driver for outputting scan signals to the scan lines and fan-out lines connecting the data lines with the display driving circuit **320** may be disposed in the non-display area NDA of the display layer DISL.

[0065] The encapsulation layer ENC may be disposed on the display layer DISL. The encapsulation layer ENC may be a layer for encapsulating the light emitting element layer of the display layer DISL to prevent oxygen or moisture from being permeated into the light emitting element layer of the display layer DISL. The encapsulation layer ENC may be disposed on an upper surface and sides of the display layer DISL.

[0066] The sensor electrode layer SENL may be disposed on the display layer DISL. The sensor electrode layer SENL may include sensor electrodes. The sensor electrode layer SENL may sense a touch through use of the sensor electrodes.

[0067] The polarizing film PF may be disposed on the sensor electrode layer SENL. The polarizing film PF may include a first base member, a linear polarizing plate, a phase delay film such as a λ/4 plate (quarter-wave plate), and a second base member. The first base member, the phase delay film, the linear polarizing plate and the second base member may be sequentially stacked on the sensor electrode layer SENL.

[0068] The cover window CW may be disposed on the polarizing film PF. The cover window CW may be attached onto the polarizing film PF by a transparent adhesive member such as an optically clear adhesive (OCA).

[0069] The panel lower cover PB may be disposed below the display panel **300**. The panel lower cover PB may be attached to a lower surface of the display panel **300** through an adhesive member. The adhesive member may be a pressure sensitive adhesive (PSA). The panel lower cover PB may include at least one of a light blocking member for absorbing light incident from the outside, a buffer member for absorbing an impact from the outside or a heat dissipation member for efficiently discharging heat of the display panel **300**.

[0070] The light blocking member may be disposed below the display panel **300**. The light blocking member blocks transmission of light to prevent components disposed below the light blocking member, for example, the display circuit board **310**, etc. from being viewed from an upper portion of the display panel **300**. The light blocking member may include a light absorbing material such as a black pigment or a black dye.

[0071] The buffer member may be disposed below the light blocking member. The buffer member absorbs an external impact to prevent the display panel **300** from being damaged. The buffer member may include a single layer or a plurality of layers. For example, the buffer member may be formed of a polymer resin such as polyurethane, polycarbonate, polypropylene and polyethylene, or may include a material having elasticity, such as a sponge formed by foaming a rubber, a urethane-based material or an acrylic-based material.

[0072] The heat dissipation member may be disposed below the buffer member. The heat dissipation member may include a first heat dissipation layer including graphite or carbon nanotubes and a second heat dissipation layer formed of a metal thin film such as copper, nickel, ferrite or silver, which may shield electromagnetic waves and have excellent thermal conductivity.

[0073] According to an embodiment, as shown in FIG. **4**, the substrate SUB may be bent in the sub-area SBA, and may be disposed below the display panel **300**. The sub-area SBA of the substrate SUB may be attached to a lower surface of the panel lower cover PB by a first adhesive member **391**. The first adhesive member **391** may be a pressure sensitive adhesive.

[0074] According to an embodiment, as shown in FIG. **4**, the protruded area PA of the substrate SUB may be bent, and may be disposed below the display panel **300**. The protruded area PA of the substrate SUB may be attached to the lower surface of the panel lower cover PB by a second adhesive member **392**. The second adhesive member **392** may be a pressure sensitive adhesive.

[0075] The display circuit board **310** may be attached to the display pads DPD of the sub-area SBA of the substrate SUB by using a conductive adhesive member such as an anisotropic conductive film and an anisotropic conductive adhesive. The display circuit board **310** may include a connector **311** connected to the flexible circuit board **312**. The display circuit board **310** may be connected to a connector **352** of the main circuit board **400** by the flexible circuit board **312**.

[0076] The touch driving circuit **330** may be disposed on the display circuit board **310**. The touch driving circuit **330** may generate touch data in accordance with a change in an electrical signal sensed in each of the sensor electrodes of the sensor electrode layer of the display panel **300** and transmit the touch data to the main processor **410** of the main circuit board **400**, and the main processor **410** may calculate touch coordinates in which the touch has occurred, by analyzing the touch data.

[0077] The antenna circuit board **340** may be attached to the antenna pads APD of the protruded area PA of the substrate SUB by using a conductive adhesive member such as an anisotropic conductive film and an anisotropic conductive adhesive. The connector **351** of the antenna circuit board **340** may be connected to the connector **351** of the main circuit board **400**. The protruded area PA may be connected to the main circuit board **400** by the antenna circuit board **340**.

[0078] The main circuit board **400** may be a rigid printed circuit board (PCB) that is rigid and thus is not easily bent. The main processor **410** and the antenna driving circuit **350** may be disposed on the main circuit board **400**.

[0079] The antenna driving circuit **350** may be electrically connected to antennas (e.g., ANT**1**, ANT**2** and ANT**3** of FIG. **7**, exemplified in detail in FIG. **13**) of the display panel **300** through the antenna circuit board **340**. The antenna circuit board **340** may be formed with an integrated circuit (IC).

[0080] The antenna driving circuit **350** may recognize a gesture of a user's hand by processing electromagnetic wave signals transmitted and received through antennas ANT. The antenna driving circuit **350** may sense biometric information such as a user's breathing and heart rate by processing electromagnetic wave signals transmitted and received through the antennas ANT.

[0081] FIG. **5** is a plan view illustrating a display device according to another embodiment. FIG. **6** is a plan view illustrating a display device according to still another embodiment.

[0082] The embodiment of FIG. **5** is different from the embodiments of FIGS. **1** and **2** in that the protruded area PA is protruded from the left side of the main area MA in the Y direction. The embodiment of FIG. **6** is different from the embodiments of FIGS. **1** and **2** in that the protruded area PA is protruded from the right side of the main area MA in the Y direction. In FIGS. **5** to **6**, the description repeated with the embodiments of FIGS. **1** and **2** will be omitted.

[0083] As shown in FIGS. **5** to **6**, the protruded area PA may be protruded from one side of the main area MA, and one side of the main area MA may be any one of an upper side, a lower side, a left side and a right side of the main area MA.

[0084] Meanwhile, although not shown, the protruded area PA may be protruded from the lower side of the main area MA in the Y direction, and the protruded area PA may be disposed to be spaced apart from the sub-area SBA in the Y direction. In this case, a length of the protruded area PA in the X direction) may be smaller than a length of the sub-area SBA in the Y direction, but the embodiment of the present disclosure is not limited thereto.

[0085] FIG. **7** is a plan view illustrating any one corner, and a portion of one side. of a display panel according to an embodiment.

[0086] Referring to FIG. **7**, the display panel **300** according to an embodiment may include antennas ANT**1**, ANT**2** and ANT**3** serving as sensors. For example, the antennas ANT**1**, ANT**2** and ANT**3** may be multi-input multi-output (MIMO) antennas that include a plurality of transmitting antennas ANT**1** and a plurality of receiving antennas ANT**2** and ANT**3**. (Detailed example configurations of antennas ANT**1** and ANT**2** are described later in connection with FIG. **13**.) For example, the antennas ANT**1**, ANT**2** and ANT**3** may serve as sensors used to recognize gestures of a user's hand or to sense biometric information such as a user's breathing and heart rate. In this case, the transmitting antenna ANT**1** may transmit a radar signal, which is reflected off the user's hand or other body part, and the reflected signal is sensed by the receiving antennas ANT**2** and ANT**3**. Note that in other embodiments, the antennas ANT**1**, ANT**2** and ANT**3** may additionally or alternatively serve other functions, such as exchanging communication signals with other electronic devices.

[0087] According to an embodiment, the antennas ANT**1**, ANT**2** and ANT**3** may be adjacent to at least one corner of the display panel **300**, and may be disposed at least partly in the non-display area NDA. (In the embodiment of FIG. **7**, antennas ANT**1** and ANT**2** have dipole arms entirely in the non-display area NDA**1**. In other embodiments, not illustrated, the antennas ANT**1** and ANT**2** have dipole arms partly located within the non-display area NDA**1** and partly in the protruded area PA. In still other embodiments, not illustrated, the antennas ANT**1** and ANT**2** are disposed entirely in the protruded area PA.) For example, as shown in FIG. **6**, the non-display area ("region") NDA may include a first non-display area NDA**1** positioned on a long side of the display panel **300**, a second non-display area NDA**2** positioned on a short side of the display panel **300**, and a third non-display area NDA**3** connecting the first non-display area NDA**1** with the second non-display area NDA**2**. Also, as shown in FIG. **6**, the display panel **300** may have a rectangular shape when viewed on a plane, and may include a first corner **601**, a second corner **602**, a third corner **603** and a fourth corner **604**. Although FIG. **7** will be described based on the first corner **601** shown in FIG. **6**, the elements included in FIG. **7** may be disposed in at least one of the second corner **602**, the third corner **603** or the fourth corner **604** of the display panel **300**.

[0088] According to an embodiment, the display panel **300** includes at least one transmitting antenna ANT**1** disposed in the first non-display area NDA**1**. For example, the transmitting antenna ANT**1** may be disposed in the first non-display area NDA**1** adjacent to any one corner of the display panel **300**. In the present disclosure, a single transmitting (Tx) antenna ANT**1** is described, but a plurality of transmitting antennas ANT**1** may be included in other examples.

[0089] According to an embodiment, the display panel **300** includes a first receiving antenna ANT**2** including a plurality of antennas, e.g., Rx**1** and Rx**2**, forming a two (or higher) element array, and disposed in the first non-display area NDA**1**. For example, the antennas Rx**1** and Rx**2** may be collectively used to steer a receive path beam. The first receiving antenna ANT**2** may be disposed in the first non-display area NDA**1** adjacent to any one corner of the display panel **300**, and may be adjacent to the transmitting antenna ANT**1**. In the present disclosure, it is described that the first receiving antenna ANT**2** includes two receiving antennas Rx**1** and Rx**2**, but the present disclosure is not limited thereto.

[0090] A distance in the X direction between the transmitting antenna ANT**1** and the closest antenna (e.g., Rx**1**) of the first receiving antenna ANT**2** may be a first distance **701**. For example, the first distance **701** may be about 0.5λ (half a wavelength in free space) at a frequency of about 60 GHz.

[0091] The distance between the antennas Rx**1** and Rx**2** may be a second distance **702**. For example, the second distance **702** may be about 0.5λ.

[0092] According to an embodiment, the display panel **300** includes a second receiving antenna ANT**3** disposed in the third non-display area NDA**3**. For example, the second receiving antenna ANT**3** is disposed in the third non-display area NDA**3** corresponding to any one corner of the display panel **300**. Unlike the plurality of antennas Rx**1** and Rx**2**, the second receiving antenna ANT**3** is oriented offset at a designated angle (e.g., about 55 degrees) from the transmitting antenna ANT**1** on a plane. For instance, the antenna ANT**3** is a monopole, and an axis running through a rod of the monopole intersects the X axis (running through dipole arms of the first transmitting antenna ANT**1**) at angle of about 55 degrees.

[0093] The plurality of first receiving antennas ANT**2** and the second receiving antenna ANT**3** may be spaced apart from each other by a third distance **703** in the X direction, and may be spaced apart from each other by a fourth distance **704** in the Y direction. For example, the third distance **703** may be about 1.2λ, and the fourth distance **704** may be about 0.5λ.

[0094] The at least one transmitting antenna ANT**1** may be referred to herein interchangeably as a first antenna ANT**1**, and the following description will discuss the case of a single transmitting antenna ANT**1** for ease of description. The first receiving antenna ANT**2** may be a two element array including the antenna Rx**1** (hereafter, "second antenna") and the antenna Rx**2** (hereafter, "third antenna"), which are oriented parallel to each other. (A common axis may run through dipole arms of each of the first antenna ANT**1**, the second antenna Rx**1** and the third antenna Rx**2**.) The first antenna ANT**1**, the second antenna Rx**1** and the third antenna Rx**2** may be of the same type and have the same shape. For example, as noted above, the first antenna ANT**1**, the second antenna Rx**1** and the third antenna Rx**2** may be slotted dipole antennas.

[0095] According to an embodiment, the second receiving antenna ANT**3** is or includes a fourth antenna Rx**3**, and a shape of the second receiving antenna ANT**3** may be different from that of the first antenna ANT**1**, the second antenna Rx**1** and the third antenna Rx**2**. For example, the fourth antenna Rx**3**, which is the second receiving antenna ANT**3**, may be a monopole antenna.

[0096] According to an embodiment, the first antenna ANT**1**, the second antenna Rx**1** and the third antenna Rx**2** are asymmetric dipole antennas having an asymmetric shape (with slotted dipole arms having different lengths) when viewed on a plane of the display panel **300**. As described above, the example of the first antenna ANT**1**, the second antenna Rx**1** and the third antenna Rx**2** being asymmetric dipole antennas allows for the design of the polarization direction of the first antenna ANT**1**, the second antenna Rx**1** and the third antenna Rx**2** to be the same as the polarization direction of the fourth antenna Rx**3**. For example, as described later with reference to FIG. **12**, the transmitting antenna ANT**1**, the receiving antenna ANT**2** and the second receiving antenna ANT**3** of the present disclosure may have substantially the same polarization direction.

[0097] When the transmitting antenna ANT**1**, the receiving antenna ANT**2** and the second receiving antenna ANT**3**, which are shown in FIG. **7**, are defined as one antenna group ANT**1**, ANT**2** and ANT**3**, the antenna group ANT**1**, ANT**2** and ANT**3** may be disposed at one corner or a plurality of corners, which is (are) selected from the corners (**601**, **602**, **603** and **604** of FIG. **6**) included in the display panel **300**. Thus, there may be a plurality of antenna groups, each including a set of antennas ANT**1**, ANT**2** and ANT**3**, according to the present disclosure.

[0098] FIG. **8** is a conceptual view illustrating operations of antennas according to one embodiment.

[0099] FIGS. **9**A and **9**B are radar images obtained by sensing an object, which is moving in a first direction, by antennas according to one embodiment.

[0100] FIGS. **10**A and **10**B are radar images obtained by sensing an object, which is moving in a second direction, by antennas according to one embodiment.

[0101] FIGS. **11**A and **11**B are radar images obtained by sensing an object, which is moving in a third direction, by antennas according to one embodiment.

[0102] For example, FIG. **8** is a conceptual view illustrating an experimental condition of radar images shown in FIGS. **9**A to **11**B. Referring to FIG. **8**, the experimental condition is set so that the object moves in a first direction DR**1** (e.g., Azimuth direction) while being spaced apart from the antennas in a normal direction (Z-direction) of the display panel **300** or moves in a second direction DR**2** (e.g., Elevation direction) or moves in a third direction DR**3** (e.g., Azimuth+Elevation direction).

[0103] In FIGS. **9**A to **11**B, an image pattern extended in a specific direction represents movement of an object sensed based on a phase difference of signals sensed from different receiving antennas.

[0104] Hereinafter, a result of sensing movement of an object by antennas according to one embodiment will be described with reference to FIGS. **8** to **11**B.

[0105] Referring to FIGS. **8**, **9**A and **9**B, the experimental condition is set so that the object moves in the first direction (e.g., Azimuth direction) while being spaced apart from the antennas in the normal direction (Z-direction) of the display panel **300**. In this case, as shown in FIG. **9**A, it may be seen that movement of the object from about 0° to about 40° is sensed due to a phase difference between the second and third antennas Rx**1** and Rx**2** disposed in parallel with each other. On the other hand, as shown in FIG. **9**B, it may be seen that movement of the object is not sensed due to no phase difference between the third and fourth antennas Rx**2** and Rx**3**.

[0106] Referring to FIGS. **8**, **10***a*A and **10**B, the experimental condition is set so that the object moves in the second direction (e.g., Elevation direction) while being spaced apart from the antennas in the normal direction (Z-direction) of the display panel **300**. In this case, as shown in FIG. **10**A, it may be seen that movement of the object is not sensed due to no phase difference between the second and third antennas Rx**1** and Rx**2** disposed in parallel with each other. On the other hand, as shown in FIG. **10**B, it may be seen that movement of the object from about 0° to about 40° is sensed due to a phase difference between the third and fourth antennas Rx**2** and Rx**3**.

[0107] Referring to FIGS. **8**, **11**A and **11**B, the experimental condition is set so that the object moves in the third direction (e.g., Azimuth+Elevation direction) while being spaced apart from the antennas in the normal direction (Z-direction) of the display panel **300**. In this case, as shown in FIG. **11**A, it may be seen that movement of the object is sensed due to a phase difference between the second and third antennas Rx**1** and Rx**2** disposed in parallel with each other. Likewise, as shown in FIG. **11**B, it may be seen that movement of the object from about 0° to about 30° is sensed due to a phase difference between the third and fourth antennas Rx**2** and Rx**3**.

[0108] As described with reference to FIGS. **8** to **11**B, it may be seen that the antenna groups ANT**1**, ANT**2** and ANT**3** according to one embodiment may effectively predict various moving directions of an external object.

[0109] FIG. **12** is a result of an experiment for polarization directions of antennas according to one embodiment.

[0110] Referring to FIG. **12**, the first antenna ANT**1**, the second antenna Rx**1** and the third antenna Rx**2** according to an embodiment are asymmetric dipole antennas (with dipole arms of different lengths) having an asymmetric shape when viewed on a plane of the display panel **300**. In this way, the reason why the first antenna ANT**1**, the second antenna Rx**1** and the third antenna Rx**2** are asymmetric dipole antennas is to design the polarization direction of the first antenna ANT**1**, the second antenna Rx**1** and the third antenna Rx**2** to be the same as the polarization direction of the fourth antenna Rx**3** that is the second receiving antenna ANT**3**. For example, unlike the remaining antennas ANT**1** and ANT**2**, the second receiving antenna ANT**3** is disposed on a corner of the display panel **300**, and thus may have an acute polarization direction.

[0111] In an embodiment of the present disclosure, the first antenna ANT**1**, the second antenna Rx**1** and the third antenna Rx**2** are designed as asymmetric dipole antennas such that they have the same polarization direction as that of the second receiving antenna ANT**3**. For example, as shown in FIG. **12**, the fourth antenna Rx**3** has a monopole antenna shape, is fed from a coplanar waveguide (CPW) transmission line and may have a polarization direction of about 46° as it is positioned on the corner of the display panel **300**. In consideration of the polarization direction of the fourth antenna Rx**3**, the first antenna ANT**1**, the second antenna Rx**1** and the third antenna Rx**2** are designed as asymmetric dipole antennas. Therefore, the first antenna ANT**1**, the second antenna Rx**1** and the third antenna Rx**2** may have asymmetric slots to have a polarization direction of about 46°. According to one embodiment of the present disclosure, the polarization direction of the first antenna ANT**1**, the second antenna Rx**1** and the third antenna Rx**2** may be designed to be the same as the polarization direction of the fourth antenna Rx**3** that is the second receiving antenna ANT**3**, whereby polarization loss may be reduced, and antenna efficiency may be increased.

[0112] FIG. **13** is a plan view illustrating the transmitting antenna ANT**1** and the first receiving antenna ANT**2** according to an embodiment. FIG. **13** schematically illustrates a first transmitting antenna ANT**1** and any one (e.g., the second antenna Rx**1**) of the antennas of the first receiving antenna ANT**2**. Since the antennas Rx**1** and Rx**2** may have the same shape and dimensions, the description of the second antenna Rx**1** may apply equally to the third antenna Rx**2**. In FIG. **13**, shaded areas represent conductive material, e.g., metal, and white areas between the shaded areas represent air or dielectric.

[0113] Referring to FIG. **13**, each of the first (transmitting) antenna ANT**1** the second antenna Rx**1** and the third antenna Rx**2** (together forming the first receiving antenna ANT**2**) may have an antenna feed extending from a coplanar waveguide CPW. The coplanar waveguide CPW may include a signal conductor s**1** (interchangeably, "feeding line") separated on opposite sides by ground conductors g**1** and g**2**. The antenna feed includes an antenna electrode AE**1** separated on opposite sides by slots SL**1** and SL**2** to extended sections (in the Y direction) of the ground conductors g**1** and g**2**.

[0114] Thus, the antenna electrode AE**1** extends the feeding line s**1**. Returning momentarily to FIG. **7**, a similar feeding line may extend from the protruded area to the right side corner of the display panel **300** to feed a transmission line (e.g., also coplanar waveguide) leading to the second receiving antenna ANT**3**.

[0115] As illustrated in FIG. **13**, each of the first to third antennas ANT**1**, Rx**1** and Rx**2** may be slotted dipole antennas, with a dipole forming slot SL**3** in the upper (radiating) portion of the dipole antenna. With the antenna electrode AE**1** as a reference axis, since the left-side slot distance (to the slot edge abutting a conductive section AE**2**) is different from the right-side slot distance (to the slot edge abutting a conductive section AE**3**), the resulting dipole is an asymmetric slotted dipole. (The conductive section AE**2** in conjunction with the left side of the slot SL**3** may be referred to herein as a second antenna electrode of a slotted dipole antenna. The conductive section AE**3** in conjunction with the right side of the slot SL**3** may be referred to herein as a third antenna electrode of a slotted dipole antenna.) By designing the dipoles in this manner, the antenna boresight for each dipole may be skewed to align with the antenna boresight of the antenna ANT**3**. Further, the polarization directions of the first to third antennas ANT**1** to ANT may be the same.

[0116] The second antenna Rx**1** electrode is extended from the first antenna ANT**1** electrode in one side direction (e.g., left direction), and has a first length in one side direction. The second antenna Rx**1** electrode may include a slot.

[0117] The third antenna Rx**2** electrode is extended from the first antenna ANT**1** electrode in the other side direction (e.g., right direction) opposite to one side direction, and has a second length in the other side direction. In this case, the second length may be longer than the first length. The third antenna Rx**2** electrode may include a slot, and a length of the slot of the third antenna Rx**2** electrode may be longer than that of the slot of the second antenna Rx**1** electrode.

[0118] According to an embodiment, a first ground conductor pattern (a ground conductor that is "patterned") connected to the second antenna Rx**1** electrode is disposed in one side direction from the feeding line, wherein the first ground pattern includes a first ground etched area GE**1** disposed in parallel with the feeding line.

[0119] According to one embodiment, a second ground pattern connected to the third antenna Rx**2** electrode is disposed in the other side direction from the feeding line, wherein the second ground pattern includes a second ground etched area GE**2** disposed in parallel with the feeding line.

[0120] In the display panel **300** according to one embodiment, the ground etch areas GE**1** and GE**2** may be disposed between the adjacent antennas ANT1, Rx1 and Rx2, so that a coupling coefficient between the antennas may be reduced to about 12 dB or more.

[0121] In the description with reference to FIGS. 7 to 13, it has been described that the transmitting antenna ANT1 and the plurality of first receiving antennas ANT2 are disposed along the long side of the display panel 300, but the present disclosure is not limited thereto. In other embodiments, the transmitting antenna ANT1 and the plurality of first receiving antennas ANT2 are disposed along the short side of the display panel 300. For example, the non-display area may include a first non-display area NDA1 positioned on the short side of the display panel 300, a second non-display area NDA2 positioned on the long side of the display panel 300, a third non-display area NDA3 positioned at the corner of the display panel 300, connecting the first non-display area NDA1 with the second non-display area NDA2, and the display panel 300 may include at least one transmitting antenna ANT1 disposed at least partially in the first non-display area NDA1, a plurality of first receiving antennas ANT2 adjacent to at least one transmitting antenna ANT1 and disposed at least partially in the first non-display area NDA1 and a second receiving antenna ANT3 disposed in the third non-display area NDA3. In still other embodiments, the profile of the display device 10 is square, such that parallel first sides are orthogonal to parallel second sides; the transmitting antenna(s) ANT1 and the adjacent first receiving antenna ANT2 are disposed along any selected side of the first sides or the second sides; and the second receiving antenna ANT3 is disposed in a corner that connects the selected side to another one of the sides.

[0122] FIG. 14 is a plan view illustrating any one corner of a display panel according to an embodiment.

[0123] The embodiment of FIG. 14 differs from the embodiment of FIG. 7 in that the second receiving antenna ANT3 is disposed in the second non-display area NDA2. For example, the second receiving antenna ANT3 may not be disposed on the corner of the display panel 300 having a curved shape when viewed in a plan view, but is instead disposed on the long side or the short side of the display panel 300. (In the case of a square display device 10, the transmitting antenna ANT1 and the first receiving antenna ANT2 may be disposed along a first side, and the second receiving antenna ANT3 may be disposed along a second side, orthogonal to the first side, that connects to the first side.) Hereinafter, only features that differ from the embodiment of FIG. 14 will be described. Features in FIG. 14 not described below may be the same as the corresponding features described above with reference to FIGS. 7 to 13.

[0124] According to the embodiment of FIG. 14, the display panel 300 includes at least one transmitting antenna ANT1 disposed in the first non-display area NDA1, a plurality of first receiving antennas Rx1 and Rx3 (forming a first receiving antenna ANT2) adjacent to the at least one transmitting antenna ANT1 in the first non-display area NDA1 and a second receiving antenna ANT3 disposed in the second non-display area NDA2. Each of the antennas ANT1, Rx1 and Rx2 may be dipoles having dipole arms oriented in the first (X) direction. The second receiving antenna ANT3 may be a monopole having a rod axis oriented in the Y (second) direction.

[0125] In FIG. 14, as the second receiving antenna ANT3 is a monopole extended in the second direction, the polarization direction of the second receiving antenna ANT3 may be the second direction. Therefore, in the embodiment of FIG. 14, unlike the embodiment of FIG. 7, the transmitting antenna ANT1 and the plurality of first receiving antennas ANT2 may be designed as symmetrical dipole antennas. Thus, the first antenna ANT1 (transmitting antenna ANT1), each of the second antenna Rx1 and third antenna Rx2 (see FIG. 7) forming the first receiving antenna ANT2, are configured as dipole antennas, and may each include a symmetrical slot when viewed on a plane of the display panel 300. Therefore, the polarization direction of each of the first to third antennas (ANT1, Rx1 and Rx2) may be the same as the polarization direction of the fourth antenna (second receiving antenna) ANT3.

[0126] FIG. 15 is a cross-sectional view taken along a portion of a display area of a display panel according to one embodiment.

[0127] Referring to FIG. 15, the display panel 300 uses a substrate SUB having flexible characteristics as a base substrate.

[0128] A display layer DISL including a thin film transistor layer TFTL and a light emitting element layer EML may be disposed on one surface of the substrate SUB, an encapsulation layer ENC may be disposed on the display layer DISL, and a sensor electrode layer SENL including sensor electrodes SE may be disposed on the encapsulation layer ENC. A polarizing film PF may be disposed on the sensor electrode layer SENL, and a cover window CW may be disposed on the polarizing film PF.

[0129] The substrate SUB may include a support substrate SSUB, a first substrate SUB1, a first buffer film BF1, a second substrate SUB2 and a second buffer film BF2. The first substrate SUB1 may be disposed on the support substrate SSUB1, the first buffer film BF1 may be disposed on the first substrate SUB1, the second buffer film BF2 may be disposed on the first buffer film BF1, and the second buffer film BF2 may be disposed on the second substrate SUB2.

[0130] The support substrate SSUB may be a rigid substrate for supporting a first substrate SUB1 and a second substrate SUB2, which are flexible. The support substrate SSUB may be formed of a plastic material, such as polycarbonate (PC) and polyethylene terephthalate (PET), or glass.

[0131] The first substrate SUB1 and the second substrate SUB2 may be formed of an organic material such as an acrylic resin, an epoxy resin, a phenolic resin, a polyamide resin and a polyimide resin. The first substrate SUB1 and the second substrate SUB2 may be formed of the same organic material or different organic materials.

[0132] Each of the first buffer film BF1 and the second buffer film BF2 may be formed of an inorganic material such as a silicon nitride layer, a silicon oxynitride layer, a silicon oxide layer, a titanium oxide layer and an aluminum oxide layer. Alternatively, each of the first buffer film BF1 and the second buffer film BF2 may be formed of a multi-film in which a plurality of layers among a silicon nitride layer, a silicon oxynitride layer, a silicon oxide layer, a titanium oxide layer and an aluminum oxide layer are alternately stacked. The first buffer film BF1 and the second buffer film BF2 may be formed of the same inorganic material or different inorganic materials.

[0133] An active layer ACT including a channel area TCH, a source area TS and a drain area TD of the thin film transistor TFT may be disposed on the second buffer film BF2. The active layer ACT may include polycrystalline silicon, single crystal silicon, low-temperature polycrystalline silicon, amorphous silicon or an oxide semiconductor material. When the active layer ACT includes polycrystalline silicon or an oxide semiconductor material, the source area TS and the drain area TD in the active layer ACT may be conductive areas doped with ions to have conductivity.

[0134] A gate insulating film **130** may be formed on the active layer ACT of the thin film transistor TFT. The gate insulating film **130** may be formed of an inorganic film, for example, a silicon nitride layer, a silicon oxynitride layer, a silicon oxide layer, a titanium oxide layer or an aluminum oxide layer.

[0135] A gate electrode TG and a first capacitor electrode CAE**1** of the thin film transistor TFT may be disposed on the gate insulating film **130**. The gate electrode TG of the thin film transistor TFT may overlap the channel area TCH in the third direction DR**3** (Z-axis direction). The gate electrode TG and the first capacitor electrode CAE**1** may be formed of a single layer or multi-layer made of any one of molybdenum (Mo), aluminum (Al), chromium (Cr), gold (Au), titanium (Ti), nickel (Ni), neodymium (Nd) and copper (Cu) or their alloy.

[0136] A first interlayer insulating film **141** may be disposed on the gate electrode TG and the first capacitor electrode CAE**1**. The first interlayer insulating film **141** may be formed of an inorganic film, for example, a silicon nitride layer, a silicon oxynitride layer, a silicon oxide layer, a titanium oxide layer or an aluminum oxide layer. The first interlayer insulating film **141** may include a plurality of inorganic films.

[0137] A second capacitor electrode CAE**2** may be disposed on the first interlayer insulating film **141**. The second capacitor electrode CAE**2** may overlap the first capacitor electrode CAE**1** in the third direction DR**3** (Z-axis direction). Therefore, a capacitor Cst may be formed by the first capacitor electrode CAE**1**, the second capacitor electrode CAE**2** and an inorganic insulating dielectric film disposed between the first capacitor electrode CAE**1** and the second capacitor electrode CAE**2** to serve as a dielectric film. The second capacitor electrode CAE**2** may be formed of a single layer or multi-layer made of any one of molybdenum (Mo), aluminum (Al), chromium (Cr), gold (Au), titanium (Ti), nickel (Ni), neodymium (Nd) and copper (Cu) or their alloy.

[0138] A second interlayer insulating film **142** may be disposed on the second capacitor electrode CAE**2**. The second interlayer insulating film **142** may be formed of an inorganic film, for example, a silicon nitride layer, a silicon oxynitride layer, a silicon oxide layer, a titanium oxide layer or an aluminum oxide layer. The second interlayer insulating film **142** may include a plurality of inorganic films.

[0139] A first connection electrode CE**1** may be disposed on the second interlayer insulating film **142**. The first connection electrode CE**1** may be connected to the drain area TD through a first contact hole CT**1** passing through the gate insulating film **130**, the first interlayer insulating film **141** and the second interlayer insulating film **142**. The first connection electrode CE**1** may be formed of a single layer or multi-layer made of any one of molybdenum (Mo), aluminum (Al), chromium (Cr), gold (Au), titanium (Ti), nickel (Ni), neodymium (Nd) and copper (Cu) or their alloy.

[0140] A first organic film **160** for planarizing a step difference due to the thin film transistors TFTs may be disposed on the first connection electrode CE**1**. The first organic film **160** may be formed of an organic film such as an acrylic resin, an epoxy resin, a phenolic resin, a polyamide resin, a polyimide resin or a polyimide resin.

[0141] A second connection electrode CE**2** may be disposed on the first organic film **160**. The second connection electrode CE**2** may be connected to the first connection electrode CE**1** through a second contact hole CT**2** passing through the first organic film **160**. The second connection electrode CE**2** may be formed as a single layer or multi-layer made of any one of molybdenum (Mo), aluminum (Al), chromium (Cr), gold (Au), titanium (Ti), nickel (Ni), neodymium (Nd) and copper (Cu) or their alloy.

[0142] A second organic film **180** may be disposed on the second connection electrode CE**2**. The second organic film **180** may be formed of an organic film such as an acrylic resin, an epoxy resin, a phenolic resin, a polyamide resin and a polyimide resin.

[0143] A light emitting element layer EML is disposed on the thin film transistor layer TFTL. The light emitting element layer EML may include light emitting elements LEL and a bank **190**.

[0144] Each of the light emitting elements LEL may include a pixel electrode **171**, a light emitting layer **172** and a common electrode **173**. Each of light emission areas represents an area in which a pixel electrode **171**, a light emitting layer **172** and a common electrode **173** are sequentially stacked and holes from the pixel electrode **171** and electrons from the common electrode **173** are coupled to each other in the light emitting layer **172** to emit light. In this case, the pixel electrode **171** may be an anode electrode, and the common electrode **173** may be a cathode electrode.

[0145] The pixel electrode **171** may be formed on the second organic film **180**. The pixel electrode **171** may be connected to a first connection electrode ANDE**1** through a third contact hole CT**3** passing through the second organic film **180**.

[0146] In a top emission structure in which light is emitted in a direction of the common electrode **173** based on the light emitting layer **172**, the pixel electrode **171** may be formed of a single layer of molybdenum (Mo), titanium (Ti), copper (Cu) or aluminum (Al), or may be formed of a stacked structure (Ti/Al/Ti) of aluminum (Al) and titanium (Ti), a stacked structure (ITO/Al/ITO) of aluminum (Al) and indium tin oxide (ITO), an APC alloy, or a stacked structure (ITO/APC/ITO) of APC alloy and ITO to increase reflectance. The APC alloy is an alloy of silver (Ag), palladium (Pd) and copper (Cu).

[0147] The bank **190** may serve to define light emission areas of display pixels. To this end, the bank **190** may be formed on the second organic film **180** to expose a partial area of the pixel electrode **171**. The bank **190** may cover an edge of the pixel electrode **171**. The bank **190** may be disposed in a contact hole passing through the second organic film **180**. Therefore, a third contact hole CT**3** passing through the second organic film **180** may be filled by the bank **190**. The bank **190** may be formed of an organic film such as an acrylic resin, an epoxy resin, a phenolic resin and a polyimide resin.

[0148] A spacer **191** may be disposed on the bank **190**. The spacer **191** may serve to support a mask during a process of manufacturing the light emitting layer **172**. The spacer **191** may be formed of an organic film such as an acryl resin, an epoxy resin, a phenolic resin, a polyamide resin and a polyimide resin.

[0149] The light emitting layer **172** is formed on the pixel electrode **171**. The light emitting layer **172** may include an organic material to emit light of a predetermined color. For example, the light emitting layer **172** may include a hole transporting layer, an organic material layer and an electron transporting layer. The organic material layer may include a host and a dopant. The organic material layer may include a material that emits predetermined light, and may be formed using a phosphorescent material or a fluorescent material.

[0150] For example, the organic material layer of the light emitting layer **172** of a first light emission area for emitting light of a first color includes a host material that contains carbazole biphenyl (CBP) or 1,3-bis(carbazole-9-yl) (mCP), and may be a phosphorescent material that includes a dopant containing any one or more selected from PIQIr(acac)(bis (1-phenylisoquinoline) acetylacetonate iridium), PQIr(acac) (bis(1-phenylquinoline)acetylacetonate iridium), PQIr(tris (1-phenylquinoline)iridium) and PtOEP(octaethylporphyrin platinum). Alternatively, the organic material layer of the light emitting layer **172** of the first light emission area may be a fluorescent material that includes PBD:Eu$(DBM)_3$(Phen) or Perylene, but is not limited thereto.

[0151] The organic material layer of the light emitting layer **172** of a second light emission area for emitting light of a second color and a fourth light emission area includes a host material that includes CBP or mCP, and may be a phosphorescent material that includes a dopant material containing Ir$(ppy)_3$(fac tris(2-phenylpyridine)iridium). Alternatively, the organic material layer of the light emitting layer **172** of a second light emission area for emitting light of a second color and a fourth light emission area may be a fluorescent material that includes Alq3(tris(8-hydroxyquinolino)aluminum), but is not limited thereto.

[0152] The organic material layer of the light emitting layer **172** of a third light emission area for emitting light of a third color includes a host material that includes CBP or mCP, and may be a phosphorescent material that includes a dopant material containing (4,6-F2ppy)2Irpic or L2BD111, but is not limited thereto.

[0153] The common electrode **173** is formed on the light emitting layer **172**. The common electrode **173** may be disposed to cover the light emitting layer **172**. The common electrode **173** may be a common layer commonly formed in the light emission areas. A capping layer may be formed on the common electrode **173**.

[0154] In the top emission structure, the common electrode **173** may be formed of a transparent conductive material (TCO) such as ITO and IZO, which may transmit light, or a semi-transmissive conductive material such as magnesium (Mg), silver (Ag) or an alloy of magnesium (Mg) and silver (Ag). When the common electrode **173** is formed of a semi-transmissive conductive material, light output efficiency may be enhanced by a micro cavity.

[0155] The encapsulation layer ENC may be formed on the light emitting element layer EML. The encapsulation layer ENC may include at least one inorganic film TFE**1** or TFE**3** to prevent oxygen or moisture from being permeated into the light emitting element layer EML. The encapsulation layer ENC may include at least one organic film to protect the light emitting element layer EML from particles such as dust. For example, the encapsulation layer ENC may include a first encapsulation inorganic film TFE**1**, an encapsulation organic film TFE**2** and a second encapsulation inorganic film TFE**3**.

[0156] The first encapsulation inorganic film TFE**1** may be disposed on the common electrode **173**, the encapsulation organic film TFE**2** may be disposed on the first encapsulation inorganic film TFE**1**, and the second encapsulation inorganic film TFE**3** may be disposed on the encapsulation organic film TFE**2**. The first encapsulation inorganic film TFE**1** and the second encapsulation inorganic film TFE**3** may be formed of a multi-layer in which one or more inorganic films of a silicon nitride layer, a silicon oxynitride layer, a silicon oxide layer, a titanium oxide layer and an aluminum oxide layer are alternately stacked. The encapsulation organic film TFE**2** may be an organic film such as an acryl resin, an epoxy resin, a phenolic resin, a polyamide resin or a polyimide resin.

[0157] The sensor electrode layer SENL is disposed on the encapsulation layer ENC. The sensor electrode layer SENL may include sensor electrodes SE.

[0158] A third buffer film BF**3** may be disposed on the encapsulation layer ENC. The third buffer film BF**3** may be a layer having insulating and optical functions. The third buffer film BF**3** may include at least one inorganic film. For example, the third buffer film BF**3** may be formed of a multi-film in which one or more inorganic films among a silicon nitride layer, a silicon oxynitride layer, a silicon oxide layer, a titanium oxide layer and an aluminum oxide layer are alternately stacked. The third buffer film BF**3** may be formed by a lamination process using a flexible material, a process such as spin coating and slit die coating, which uses a solution type material, or a deposition process. The third buffer film BF**3** may be omitted.

[0159] First connection portions BE**1** may be disposed on the third buffer film BF**3**. The first connection portions BE**1** may be formed of a single layer of molybdenum (Mo), titanium (Ti), copper (Cu) or aluminum (Al), or may be formed of a stacked structure (Ti/Al/Ti) of aluminum (Al) and titanium (Ti), a stacked structure (ITO/Al/ITO) of aluminum (Al) and indium tin oxide (ITO), an APC alloy, or a stacked structure (ITO/APC/ITO) of APC alloy and ITO.

[0160] A first sensor insulating film TINS**1** may be disposed on the first connection portions BE**1**. The first sensor insulating film TINS**1** may be a layer having insulating and optical functions. The first sensor insulating film TINS**1** may be formed of an inorganic film, for example, a silicon nitride layer, a silicon oxynitride layer, a silicon oxide layer, a titanium oxide layer or an aluminum oxide layer. The first sensor insulating film TINS**1** may be formed by a lamination process using a flexible material, a process such as spin coating and slit die coating, which uses a solution type material, or a deposition process.

[0161] The sensor electrodes SE, i.e., driving electrodes TE and sensing electrodes RE may be disposed on the first sensor insulating film TNIS**1**. Also, dummy patterns (not shown) may be disposed on the first sensor insulating film TNIS**1**. The driving electrodes TE, the sensing electrodes RE and the dummy patterns do not overlap the light emission areas. The driving electrodes TE, the sensing electrodes RE and the dummy patterns may be formed of a single layer of molybdenum (Mo), titanium (Ti), copper (Cu) or aluminum (Al), or may be formed of a stacked structure (Ti/Al/Ti) of aluminum (Al) and titanium (Ti), a stacked structure (ITO/Al/ITO) of aluminum (Al) and indium tin oxide (ITO), an APC alloy, or a stacked structure (ITO/APC/ITO) of APC alloy and ITO.

[0162] A second sensor insulating film TINS**2** may be disposed on the driving electrodes TE, the sensing electrodes RE and the dummy patterns (not shown). The second sensor insulating film TINS**2** may be a layer having insulating and optical functions. The second sensor insulating film TINS2 may include at least one of an inorganic film or an organic film. The inorganic film may be a silicon nitride layer, a silicon oxynitride layer, a silicon oxide layer, a titanium oxide layer or an aluminum oxide layer. The organic film may be an acrylic resin, an epoxy resin, a phenolic resin, a polyamide resin or a polyimide resin. The second sensor insulating film TINS**2** may be formed by a lamination process using a flexible material, a process such as spin coating and slit die coating, which uses a solution type material, or a deposition process.

[0163] A heat dissipation layer HSL of a panel lower cover PB may be disposed on a lower surface of the support substrate SSUB of the substrate SUB. The heat dissipation layer HSL may be formed of a metal thin film such as copper, nickel, ferrite or silver, which may shield electromagnetic waves and has excellent thermal conductivity.

[0164] FIG. **16** is a cross-sectional view taken along a boundary between a protruded area of a display device according to one embodiment and a non-display area adjacent thereto.

[0165] Referring to FIG. **16**, dams DAM**1** and DAM**2** surrounding the display area DA may be disposed in the non-display area NDA of the display panel **300**. The dam DAM may include a first dam DAM**1** of the encapsulation layer ENC and a second dam DAM**2** disposed outside the first dam DAM**1**, but the present disclosure is not limited thereto. The first dam DAM**1** and the second dam DAM**2** may be disposed on the second interlayer insulating film **142**.

[0166] The first dam DAM**1** may include a first sub-dam SDAM**1**, a second sub-dam SDAM**2** and a third sub-dam SDAM**3**, which are sequentially stacked. The first sub-dam SDAM**1** may be formed of the same material as that of the first organic film **160**, the second sub-dam SDAM**2** may be formed of the same material as that of the second organic film **180**, and the third sub-dam SDAM**3** may be formed of the same material as that of the bank **190**.

[0167] The second dam DAM**2** may include a first sub-dam SDAM**1**, a second sub-dam SDAM**2**, a third sub-dam SDAM**3** and a fourth sub-dam SDAM**4**, which are sequentially stacked. The first sub-dam SDMA**1** may be formed of the same material as that of the first organic film **160**, and the second sub-dam SDAM**2** may be formed of the same material as that of the second organic film **180**. The third sub-dam SDAM**3** may be formed of the same material as that of the bank **190**, and the fourth sub-dam SDAM**4** may be formed of the same material as that of the spacer (**191** of FIG. **15**).

[0168] In FIG. **16**, an antenna electrode AE may refer to the antennas described with reference to FIGS. **7** to **14**. For example, the antenna electrode AE of FIG. **16** may represent at least one of the transmitting antenna ANT**1**, the first receiving antenna ANT**2** or the second receiving antenna ANT**3**, which is shown in FIG. **7**.

[0169] Referring to FIG. **16**, the antenna electrode AE may include first to eighth antenna electrode layers AEL**1** to AEL**8**, but the embodiment of the present disclosure is not limited thereto. The antenna electrode AE may include only some antenna electrode layers among the first to eighth antenna electrode layers AEL**1** to AEL**8**.

[0170] The first antenna electrode layer AEL**1** is formed by the same process as that of the gate electrode (TG of FIG. **15**) and the first capacitor electrode (CAE**1** of FIG. **15**) of the thin film transistor (TFT of FIG. **15**), and may be formed of the same material as that thereof.

[0171] The second antenna electrode layer AEL**2** may be disposed on the first antenna electrode layer AEL**1** exposed without being covered by the first interlayer insulating film (**141** of FIG. **15**). The second antenna electrode layer may be formed by the same process as that of the second capacitor electrode (CAE**2** of FIG. **15**), and may be formed of the same material as that thereof.

[0172] The third antenna electrode layer AEL**3** may be disposed on the second antenna electrode layer AEL**2** exposed without being covered by the second interlayer insulating film (**142** of FIG. **15**). The third antenna electrode layer AEL**3** may be formed by the same process as that of the first connection electrode (CE**1** of FIG. **15**), and may be formed of the same material as that thereof.

[0173] The fourth antenna electrode layer AEL**4** may be disposed on the third antenna electrode layer AEL**3**. The fourth antenna electrode layer AEL**4** may be formed by the same process as that of the second connection electrode (CE**2** of FIG. **15**), and may be formed of the same material as that thereof.

[0174] The fifth antenna electrode layer AEL**5** may be disposed on the fourth antenna electrode layer AEL**4**. The fifth antenna electrode layer AEL**5** may be formed by the same process as that of the pixel electrode (**171** of FIG. **15**), and may be formed of the same material as that thereof.

[0175] The sixth antenna electrode layer AEL**6** may be disposed on the fifth antenna electrode layer AEL**5**. The sixth antenna electrode layer AEL**6** may be formed by the same process as that of the common electrode (**173** of FIG. **15**), and may be formed of the same material as that thereof.

[0176] The seventh antenna electrode layer AEL**7** may be disposed on the sixth antenna electrode layer AEL**6**. The seventh antenna electrode layer AEL**7** may be formed by the same process as that of the first connection portion (BE**1** of FIG. **15**) of the sensor electrode layer SENL, and may be formed of the same material as that thereof.

[0177] The eighth antenna electrode layer AEL**8** may be disposed on the seventh antenna electrode layer AEL**7**. The eighth antenna electrode layer AEL**8** may be formed by the same process as that of the driving electrode (TE of FIG. **15**), the sensing electrode (RE of FIG. **15**) and/or the dummy pattern (not shown) of the sensor electrode layer SENL, and may be formed of the same material as that thereof.

[0178] A through hole (or contact hole) CT may pass through the first substrate SUB**1**, the first buffer film BF**1**, the second substrate SUB**2** and the second buffer film BF**2** of the substrate SUB. Also, the through hole CT may pass through the gate insulating film **130**.

[0179] The antenna electrode AE may be in contact with a feeding line FL through the through hole CT.

[0180] An antenna pad APD electrically connected to the feeding line FL may be disposed at an end of the feeding line FL. The feeding line FL and the antenna pad APD may be disposed on a lower surface of the first substrate SUB**1** of the substrate SUB. Since a protruded area PA is disposed to be bent below a main area MA, the support substrate SSUB of the substrate SUB may be removed from the protruded area PA in which the feeding line FL is disposed.

[0181] The antenna pad APD may be connected to an antenna circuit board **340** by using an anisotropic conductive film ACF including a conductive ball CB and a conductive adhesive member CAM such as an anisotropic conductive adhesive.

[0182] In concluding the detailed description, those skilled in the art will appreciate that many variations and modifications can be made to the example embodiments without substantially departing from the principles of the present inventive concept. Therefore, the disclosed example embodiments have been presented in a generic and descriptive sense only and not for purposes of limitation.

What is claimed is:

1. A display device comprising

a display panel including a display area and a non-display region disposed outside the display area, the display panel including a flexible substrate including a protruded area extended from a portion of the non-display region, the display panel having a long side and a short side; and

an antenna circuit board connected to the protruded area,

wherein the non-display region includes a first non-display area along the long side of the display panel, a second non-display area along the short side of the display panel and a third non-display area on a corner of the display panel, connecting the first non-display area with the second non-display area, and

the display panel includes at least one transmitting antenna disposed at least partly in or proximate to the first non-display area, a plurality of first receiving antennas disposed adjacent to the at least one transmitting antenna and at least partly in or proximate to the first non-display area, and a second receiving antenna disposed in the third non-display area.

2. The display device of claim **1**, wherein the at least one transmitting antenna includes a first antenna, and the plurality of first receiving antennas include a second antenna and a third antenna, which are oriented in parallel to one other and receive with the same polarization.

3. The display device of claim **2**, wherein the first to third antennas each have the same shape.

4. The display device of claim **3**, wherein the shape of the first to third antennas is different from a shape of the second receiving antenna.

5. The display device of claim **4**, wherein the first to third antennas are dipole antennas, and

the second receiving antenna is a monopole antenna.

6. The display device of claim **5**, wherein the first to third antennas have an asymmetric shape when viewed on a plane of the display panel.

7. The display device of claim **6**, wherein each of the first to third antennas includes:

a first antenna electrode extended from a feed line;

a second antenna electrode extended in a first side direction from the first antenna electrode, having a first length in the first side direction; and

a third antenna electrode extended from the second antenna electrode in an opposite side direction to the first side direction, having a second length in the opposite side direction different from the first length.

8. The display device of claim **7**, wherein a first ground conductive pattern connected to the second antenna electrode is disposed in the first side direction from the feeding line, the first ground pattern including a first ground etched area forming a first elongated opening disposed in parallel with the feeding line, and

a second ground pattern connected to the third antenna electrode is disposed in the opposite side direction from the feeding line, the second ground pattern including a second ground etched area forming a second elongated opening disposed in parallel with the feeding line.

9. A display device comprising:

a display panel including a display area and a non-display region disposed outside the display area, the display panel including a flexible substrate including a protruded area extended from a portion of the non-display region, the display panel having a long side and a short side; and

an antenna circuit board connected to the protruded area,

wherein the non-display region includes a first non-display area along the short side of the display panel, a second non-display area along the long side of the display panel and a third non-display area positioned on a corner of the display panel, connecting the first non-display area with the second non-display area, and

the display panel includes at least one transmitting antenna disposed at least partly in or proximate to the first non-display area, a plurality of first receiving antennas adjacent to the at least one transmitting antenna and disposed at least partly in or proximate to the first non-display area, and a second receiving antenna disposed in the third non-display area.

10. The display device of claim **9**, wherein the at least one transmitting antenna includes a first antenna, and the plurality of first receiving antennas include a second antenna and a third antenna, which are disposed in parallel with each other.

11. The display device of claim **10**, wherein the first to third antennas have the same shape.

12. The display device of claim **11**, wherein the shape of the first to third antennas is different from a shape of the second receiving antenna.

13. The display device of claim **12**, wherein the first to third antennas have an asymmetric shape when viewed on a plane of the display panel.

14. A display device comprising:

a display panel including a display area and a non-display region disposed outside the display area, the display panel including a flexible substrate including a protruded area extended from a portion of the non-display area, the display panel having a long side and a short side; and

an antenna circuit board connected to the protruded area,

wherein the non-display area includes a first non-display area along a long side of the display panel, a second non-display area along a short side of the display panel and a third non-display area positioned on a corner of the display panel, connecting the first non-display area with the second non-display area, and

the display panel includes at least one transmitting antenna disposed at least partly in or proximate to the first non-display area, a plurality of first receiving antennas disposed adjacent to the at least one transmitting antenna and at least partly in or proximate to the first non-display area, and a second receiving antenna disposed in the second non-display area.

15. The display device of claim **14**, wherein the at least one transmitting antenna includes a first antenna, and the plurality of first receiving antennas include a second antenna and a third antenna, which are disposed in parallel with each other.

16. The display device of claim **15**, wherein the first to third antennas have the same shape.

17. The display device of claim **16**, wherein the shape of the first to third antennas is different from a shape of the second receiving antenna.

18. The display device of claim **17**, wherein the first to third antennas are dipole antennas, and the second receiving antenna is a monopole antenna.

19. The display device of claim **18**, wherein the first to third antennas have a symmetric shape when viewed on a plane of the display panel.

20. The display device of claim **19**, wherein a polarization direction of each of the first to third antennas is the same as a polarization direction of the second receiving antenna.

21. A display device comprising

a display panel including a display area and a non-display region disposed outside the display area, the display panel including a flexible substrate including a protruded area extended from a portion of the non-display region, the display panel having a first side and a second side orthogonal to the first side; and

an antenna circuit board connected to the protruded area,

wherein the non-display region includes a first non-display area along the first side of the display panel, a second non-display area along the second side of the display panel and a third non-display area on a corner of the display panel, connecting the first non-display area with the second non-display area, and

the display panel includes at least one transmitting antenna disposed at least partly in or proximate to the first non-display area, a plurality of first receiving antennas disposed adjacent to the at least one transmitting antenna and at least partly in or proximate to the first non-display area, and a second receiving antenna disposed in the third non-display area.

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